Fundamentos de Aprendizagem Automática

Class 01 — Learning as Compression, Pitfalls & Taxonomies

Mário Antunes

Universidade de Aveiro

2026

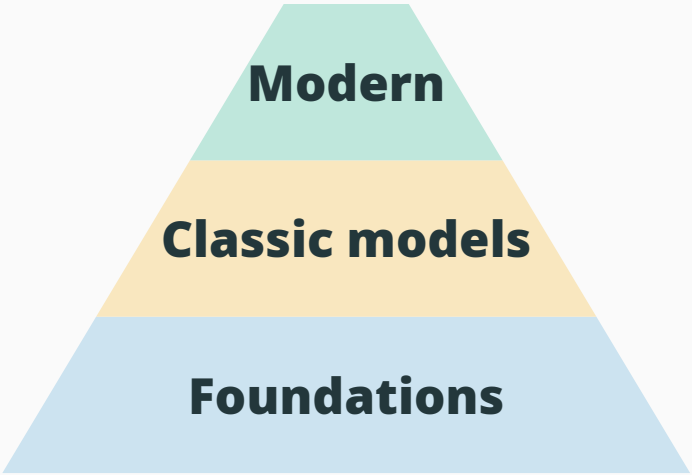
Course and Class

Professor

- **Mário Antunes**
- Associate Professor, UA
- Researcher, Instituto de Telecomunicações
- `mario.antunes@ua.pt`
- Office: 19.2.15
- Research: machine learning, time series, optimization
- Maintains the Python libraries of this course



A Fundamentals Course



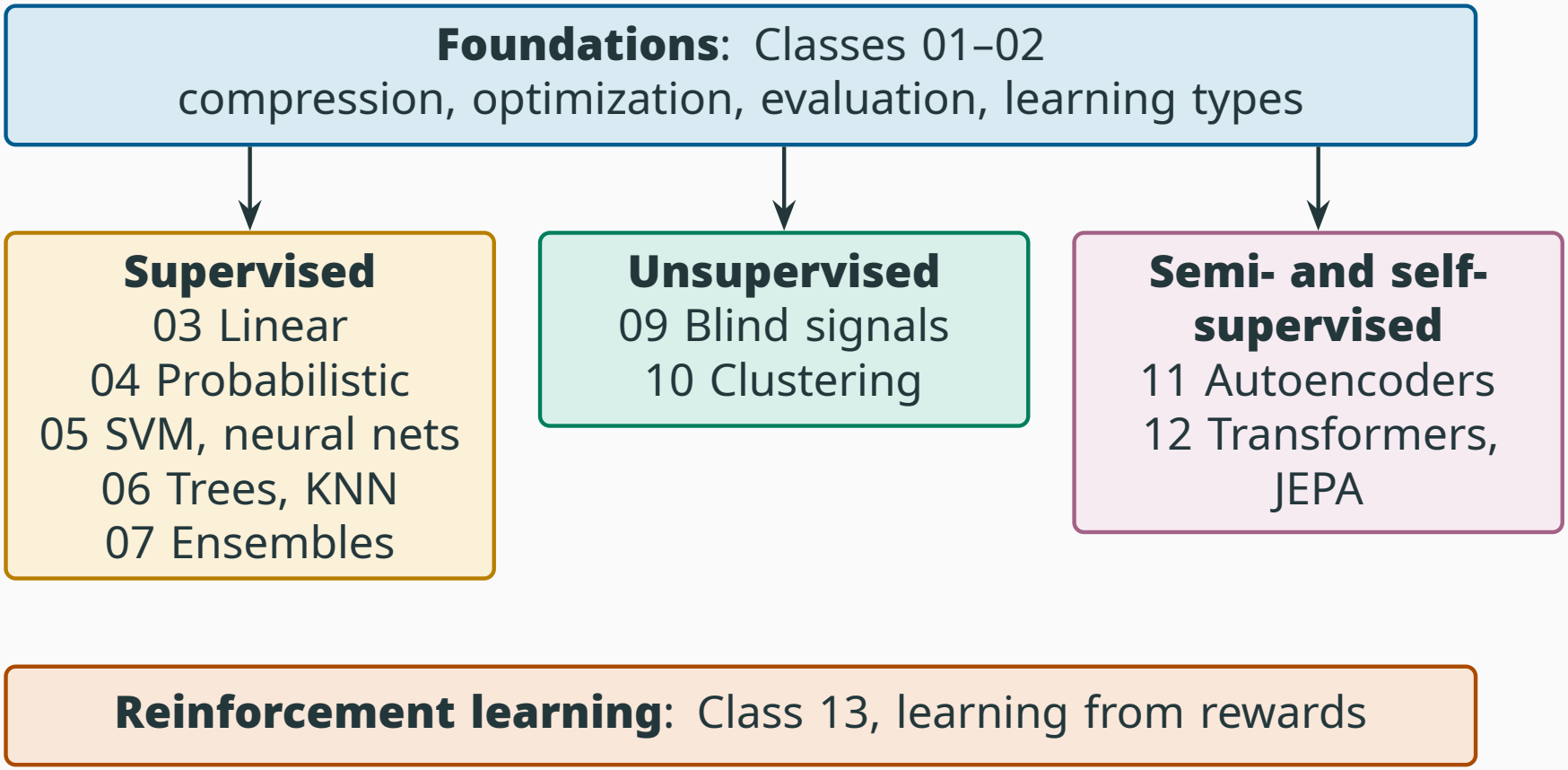
autoencoders, Transformers, JEPA

linear, SVM, neural nets,
trees, ensembles

compression, optimization, evaluation

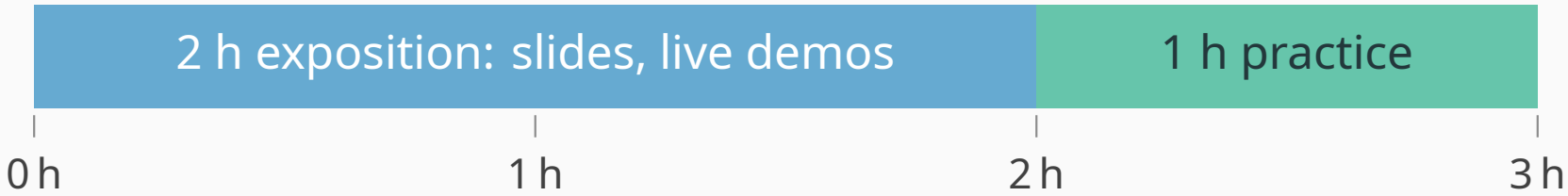
- Understand why learning works before using the latest architectures
- Every model in the course rests on the same foundations

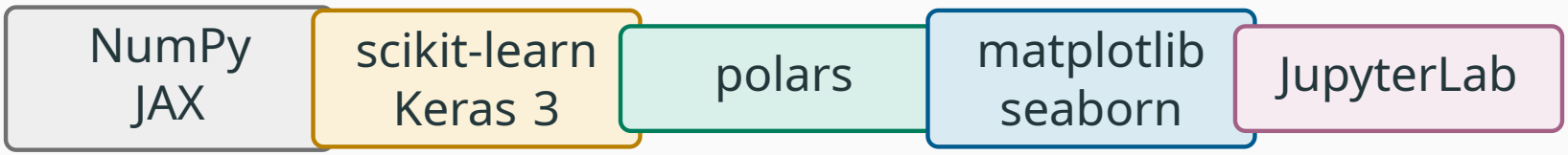
Course Topics and How They Connect



- Every later class reuses the foundations of Classes 01–02

Class Format and Environment



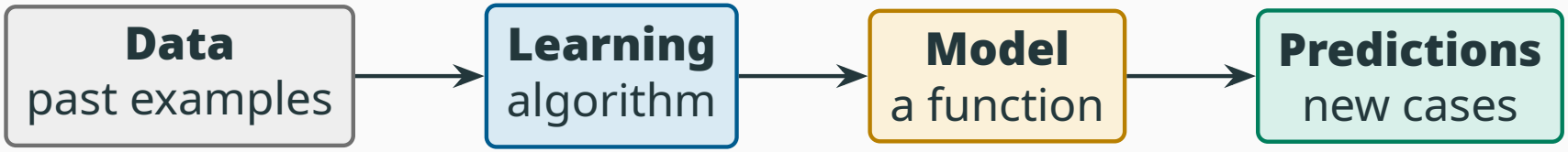



```
git clone https://github.com/detiuaveiro/faa.git && cd faa
export KERAS_BACKEND=jax    # add to ~/.bashrc
make venv && source venv/bin/activate
```

- Schedule, tests and grading: the course README, section 5

What Is Learning?

The Core Question and a Classic Definition



A program learns if its performance at a task
improves with experience (after T. Mitchell, 1997)

- It receives examples from the past
- It must act well on cases it has never seen
- Improvement needs a measure of performance

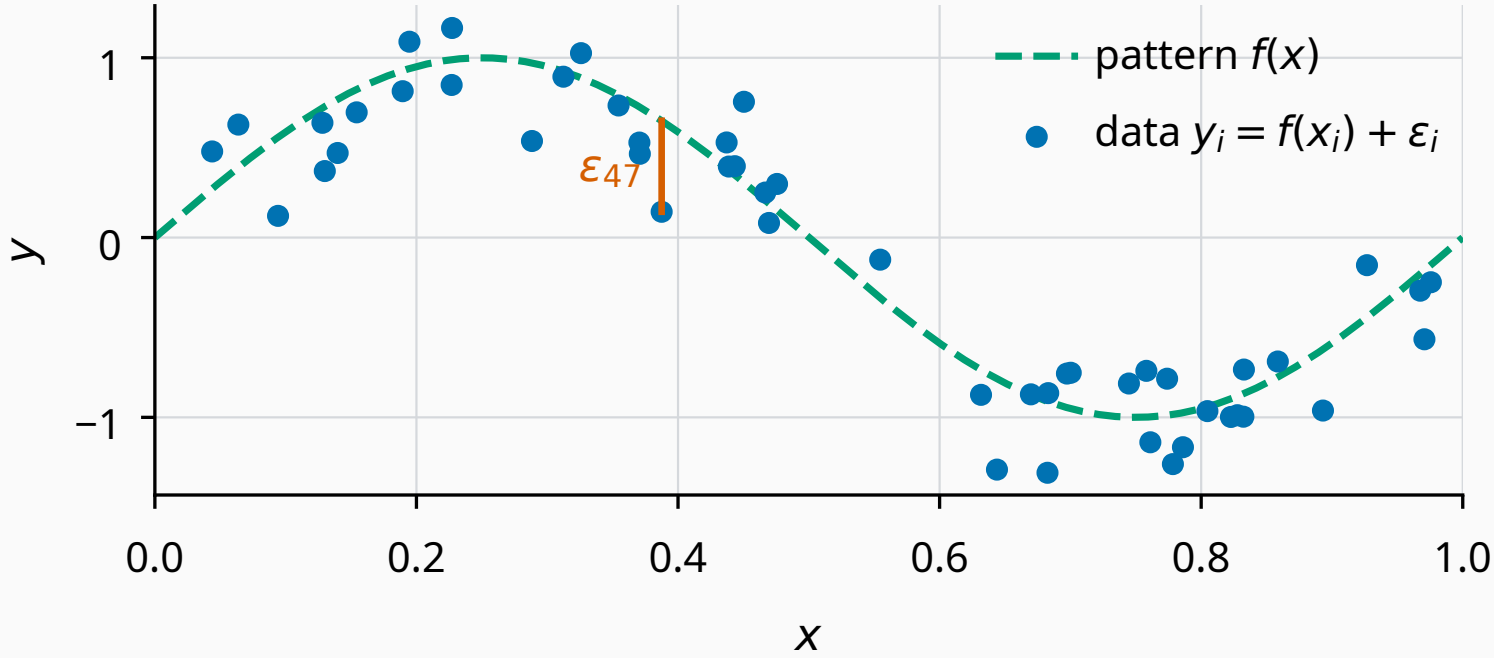
Notation N1: Data and Terms

Example 1 is a noisy sine: 50 points with one feature.

Symbol	Meaning	Example 1
n, d	number of examples, of features	50, 1
X, x_i	data matrix, example i	$x_1 = 0.774$
$y, y_i, \hat{y}_i$	targets and prediction, per example	$y_1 = -0.785$, $\hat{y}_1 = -1.004$
$f, \hat{f}$	true **pattern**, learned estimate: the **model**	$f(x_1) = -0.989$
ε, σ^2	**noise**, noise variance: the random part	$\sigma^2 = 0.09$
θ, m	parameters, their number	cubic: $m = 4$
D	dataset: the **training set** of pairs	50 pairs
$\mathcal{L}$	**loss**: one number that scores the errors	squared error

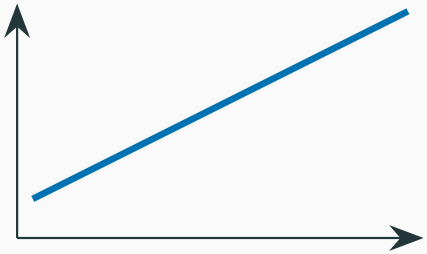
Data = Pattern + Noise

Each target is the pattern plus noise: $y_i = f(x_i) + \varepsilon_i$.

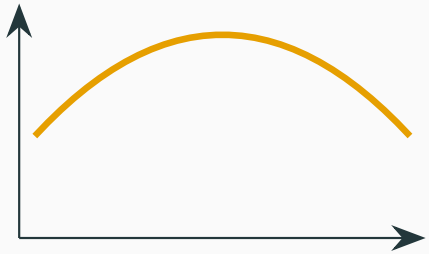


- We never see f, only the sum; the noise has $\sigma^2 = 0.09$
- Example: $-0.989 + 0.204 = -0.785$ for $i = 1$
- We see n pairs, not the whole process; the goal is new x

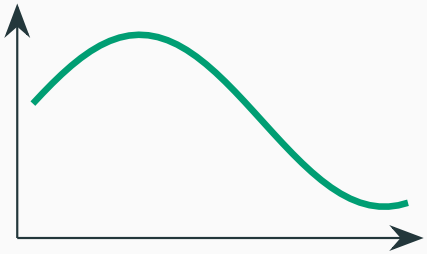
Two Models: a Line and a Polynomial



$q = 1$: a line



$q = 2$: a parabola

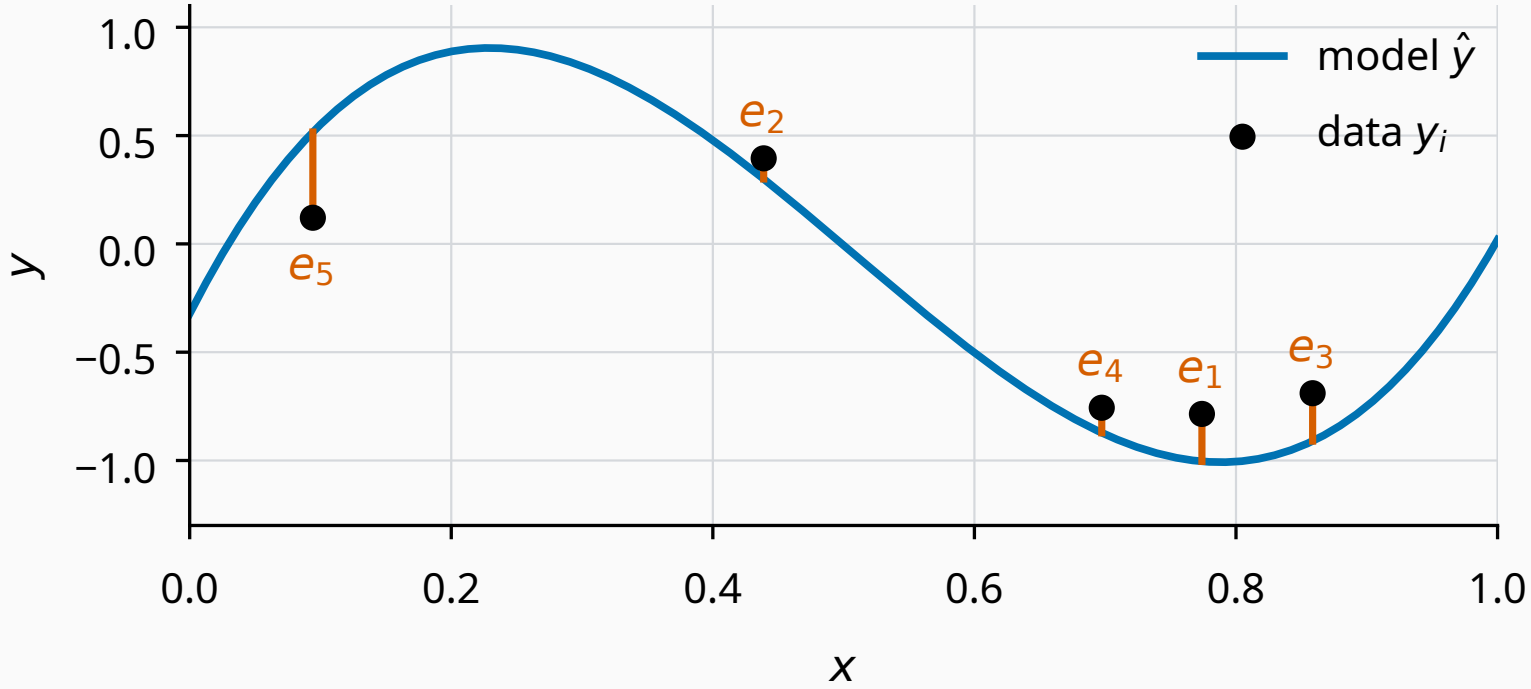


$q = 3$: a cubic

- Line: $\hat{y} = wx + b$, with $m = 2$ parameters (w, b)
- Degree q: $\hat{y} = \theta_0 + \theta_1 x + \cdots + \theta_q x^q$, $m = q + 1$
- A least-squares fit gives the cubic of Example 1 its 4 numbers: $-0.322, 11.854, -33.375, 21.862$

Errors and Mean Squared Error

Error $e_i = y_i - \hat{y}_i$; the loss is their mean square.

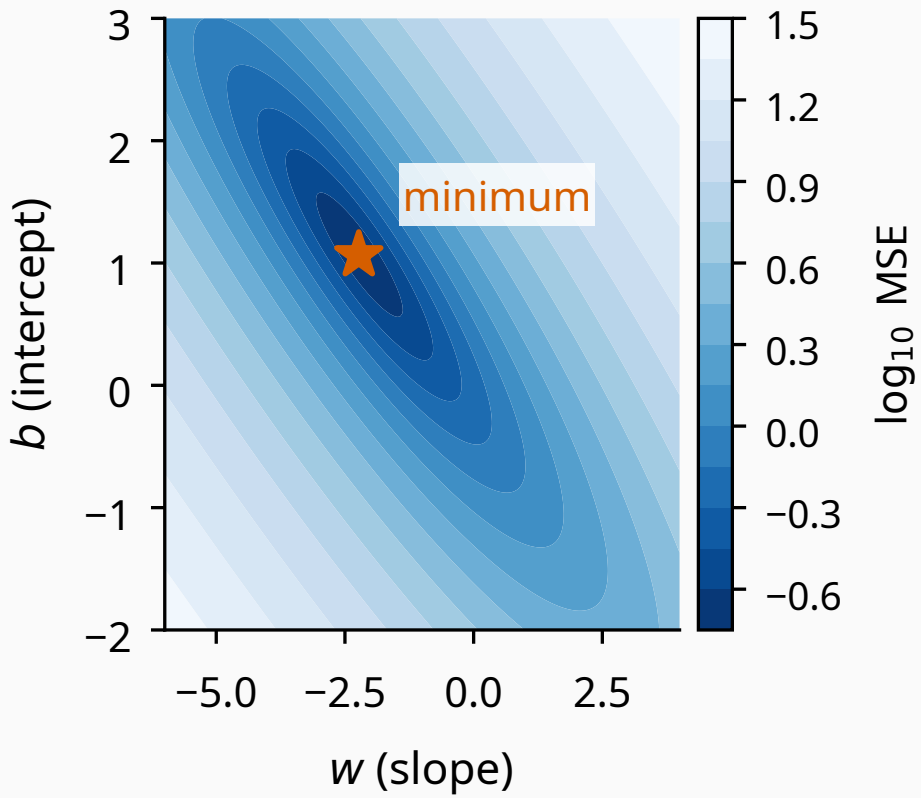


$$\mathcal{L}(\theta) = \frac{1}{n}\sum_{i=1}^{n} e_i^2 \quad \text{(definition of the MSE loss)}$$

- Five errors: sum 0.2553, squares 0.2759, loss 0.0552

Fitting Means Minimizing the Loss

Fit a line to Example 1: the loss depends on its two parameters $\theta = (w, b)$.



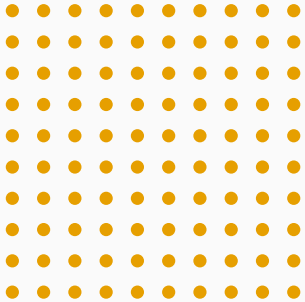
- The star is the lowest loss
- min: 0.1962
- arg min: $w = -2.219$, $b = 1.068$
- The best parameters are the fit

$$\theta^\star = \arg\min_\theta \mathcal{L}(\theta) \quad \text{(definition of the best parameters)}$$

Why Not Try Every Value?



1 parameter: g tries



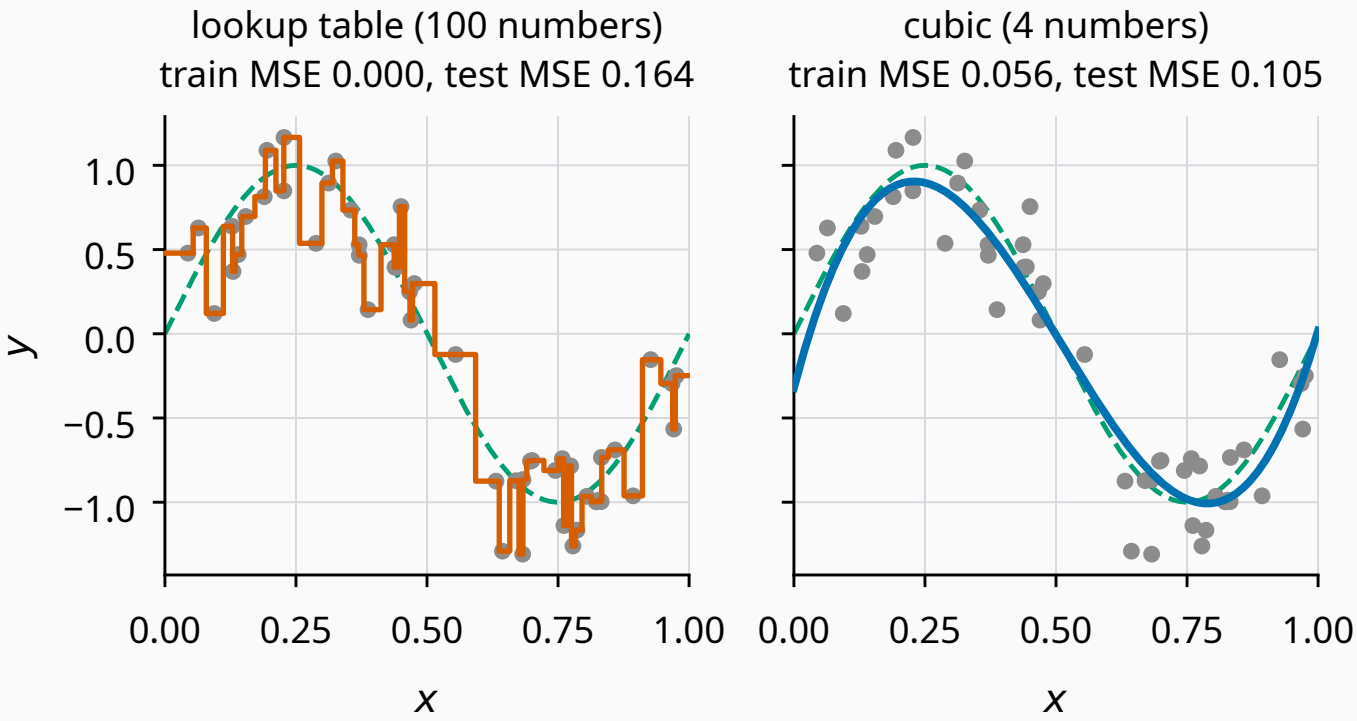
2 parameters: g^2 tries

1000 parameters:
g^{1000} tries
impossible

- A grid with g values per parameter needs g^m evaluations
- The demo grid: $g = 200$, 40000 evaluations for two parameters
- Real models have thousands to billions of parameters; Class 02 searches smarter

Memorization and Generalization

A lookup table copies the closest of 50 stored examples.



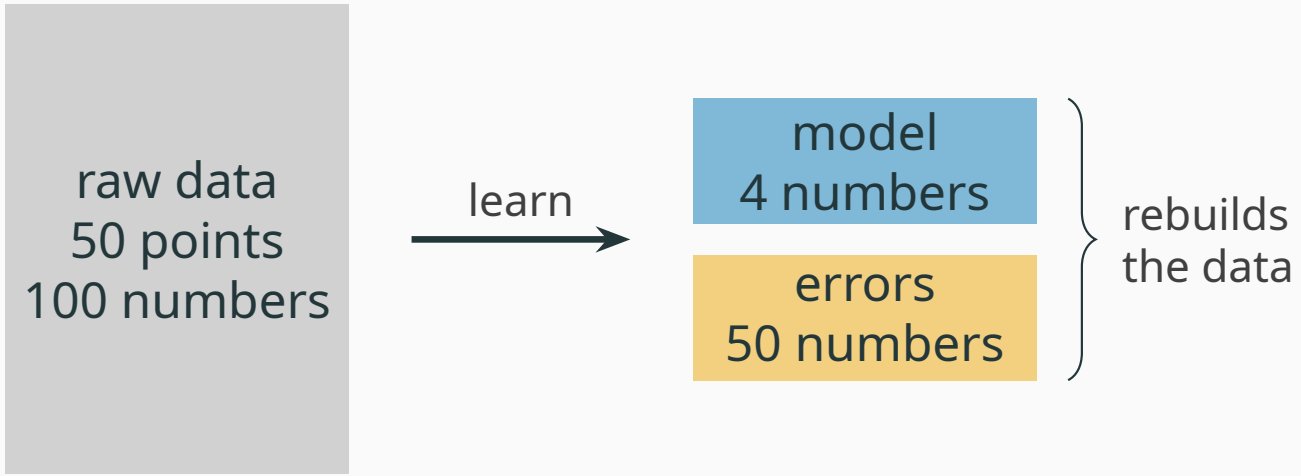
- **Memorization**: exact recall, training error 0.000
- **Generalization**: quality on 1000 new points
- Test error: table 0.164, cubic 0.105
- Rigid models miss the pattern; flexible ones copy errors

DEMO D1

`01_learning_as_compression.ipynb`

Learning as Compression

To Learn Is to Compress



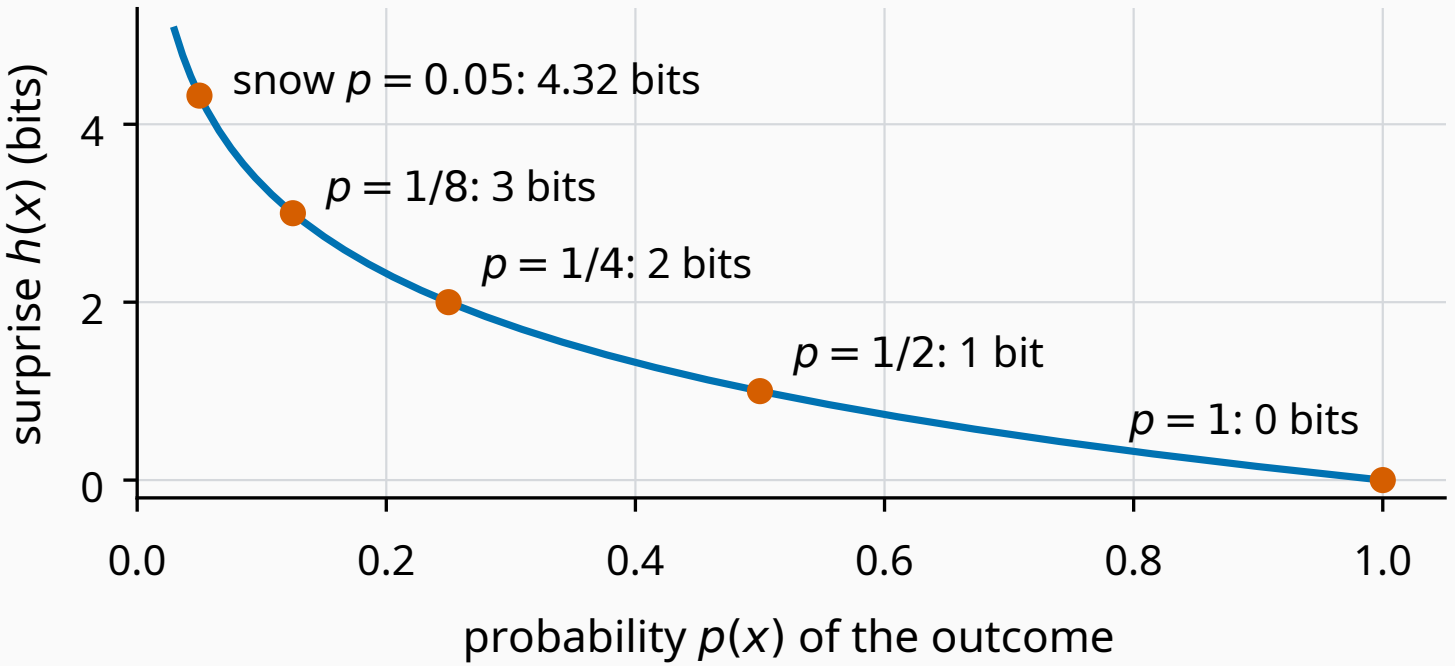
- A short model plus its errors can rebuild the data
- The model is a compressed description of the data
- A lookup table is as big as the data: nothing learned
- Pure noise has no short description: nothing to learn

Notation N2: Information

Symbol	Meaning	Example
$p(x)$	probability of outcome x (mass or density)	$p(\text{rain}) = 0.2$
$p(y \mid x)$	probability of y given x	spam given a word
$h(x)$	surprise of outcome x, in bits	$h(\text{rain}) = 2.3219$
$H(X)$	entropy: average surprise of a source X	fair coin: 1 bit
$\ell(M)$	bits to describe the model M	$q = 5$: 16.9
$\ell(D \mid M)$	bits for the data D given M (its errors)	$q = 5$: -110.4

Surprise and the Logarithm

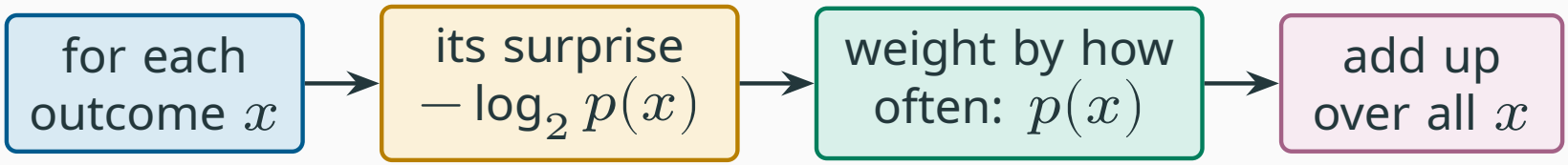
The surprise of an outcome is $h(x) = -\log_2 p(x)$ bits.



- $p(x)$ lies between 0 (never) and 1 (always)
- A rare outcome surprises more; a certain one, 0 bits
- One bit is one yes/no answer: $\log_2 8 = 3$ questions

Shannon Entropy

Entropy is the average surprise of a source: how unpredictable its next outcome is.

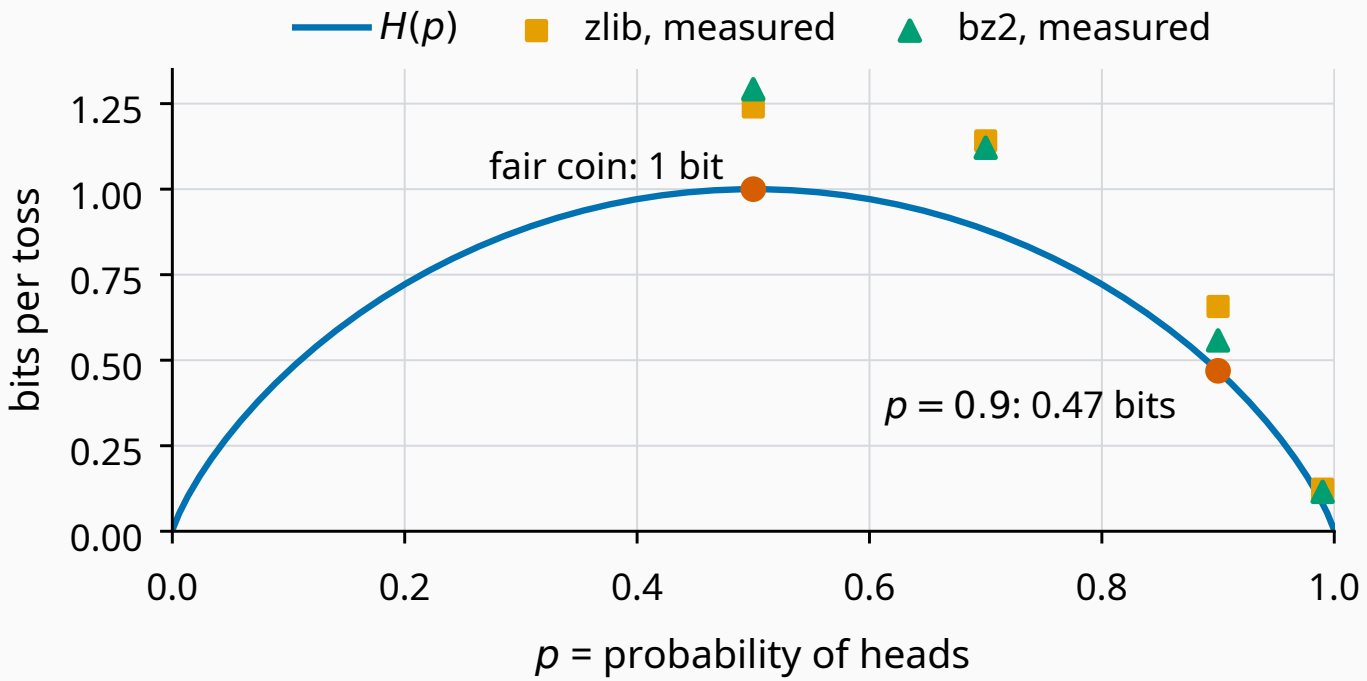


$$H(X) = -\sum_{x} p(x) \log_2 p(x) \quad \text{(definition of entropy)}$$

- High entropy: unpredictable; low entropy: predictable
- Claude Shannon introduced it in 1948

Entropy Is the Limit of Compression

No code beats the entropy on average (Shannon, 1948).



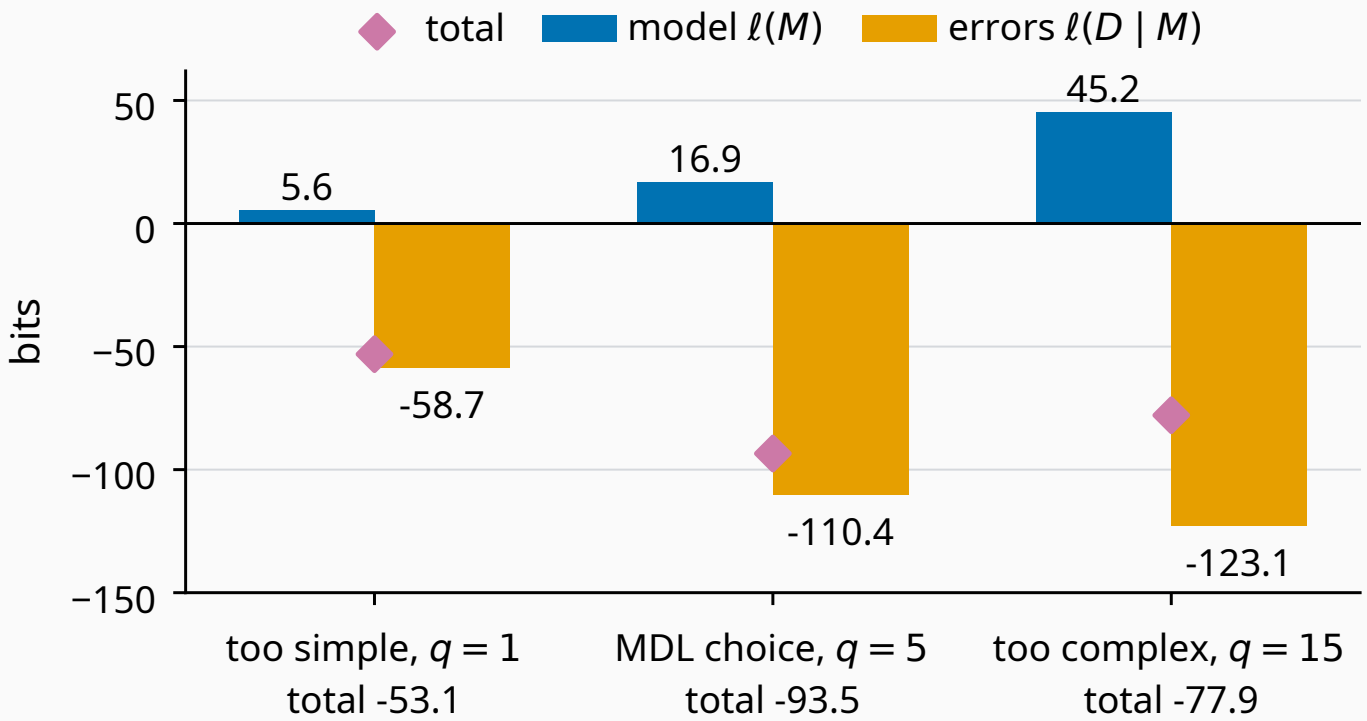
- Fair coin: $H = 1$ bit; compressors zlib, bz2 need 1.2403, 1.2921
- Biased: $H = 0.9 \times 0.152 + 0.1 \times 3.32 = 0.47$

DEMO D2

`01_learning_as_compression.ipynb`

Occam's Razor and the MDL Trade-Off

Total length: $\ell(D, M) = \ell(M) + \ell(D \mid M)$, model bits plus error bits.



- Occam's razor: among equal fits, prefer the simplest
- Shortest total wins (Rissanen, 1978)
- Constants are dropped: a length can be negative

MDL in Practice: Gaussian Errors

For Gaussian (bell-shaped) errors the two lengths have closed forms. With m parameters and n examples:

model cost $\ell(M)$
each parameter costs $(1/2)\log_2 n$ bits; grows with m

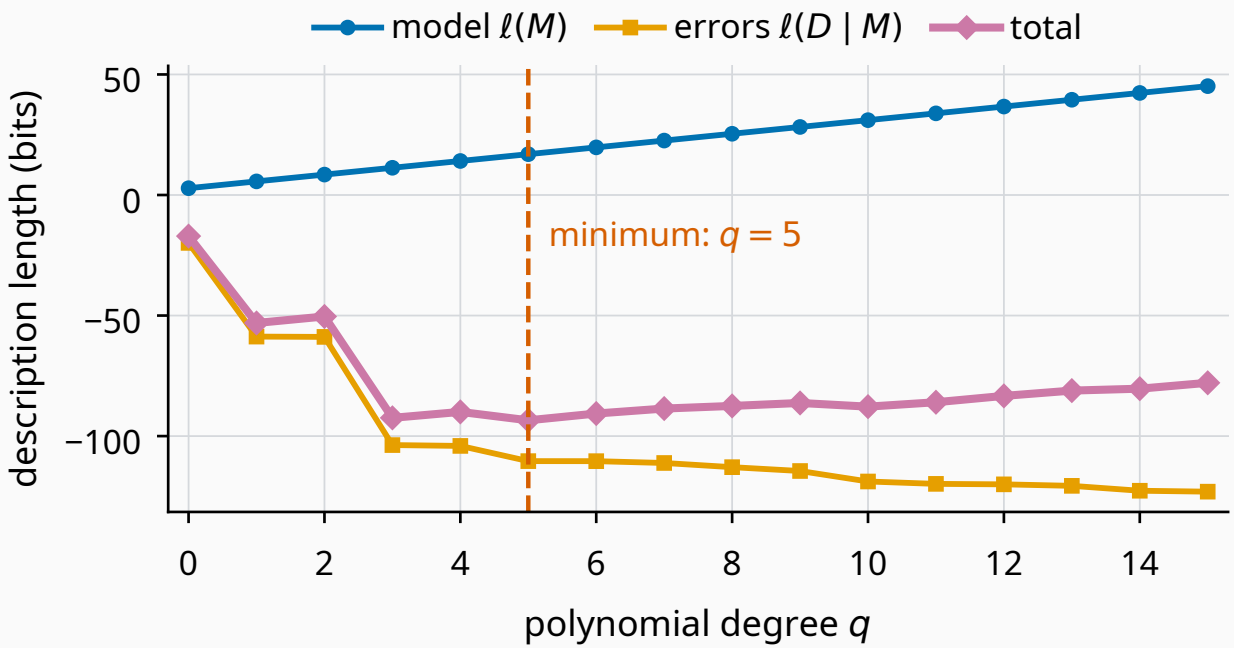
error cost $\ell(D \mid M)$
$(n/2)\log_2$ MSE bits
shrinks as the fit improves

$$\ell(M) + \ell(D \mid M) = \frac{m}{2}\log_2 n + \frac{n}{2}\log_2 \text{MSE} \quad \text{(by Rissanen)}$$

- This is the Bayesian information criterion (BIC), in bits
- It needs only the training data; the derivation is in the appendix

Worked Example: Choosing a Degree

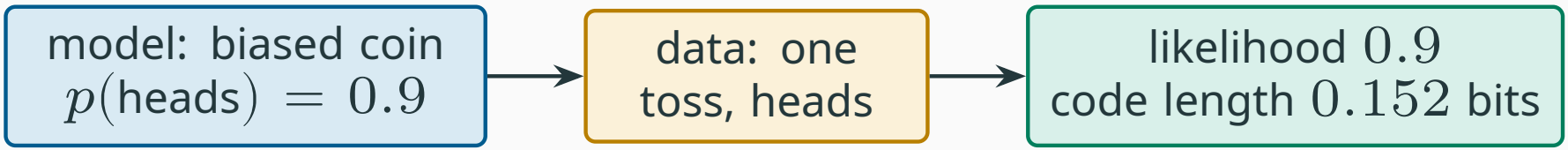
Example 1, $n = 50$: total length for each degree q.



- Minimum $q = 5$: -93.5 bits; $q = 3$: -92.4, one bit more
- Over 20 repeated draws the choice runs from 3 to 7
- With 15 points it picks $q = 14$: it fails

Likelihood: How Probable Are the Data?

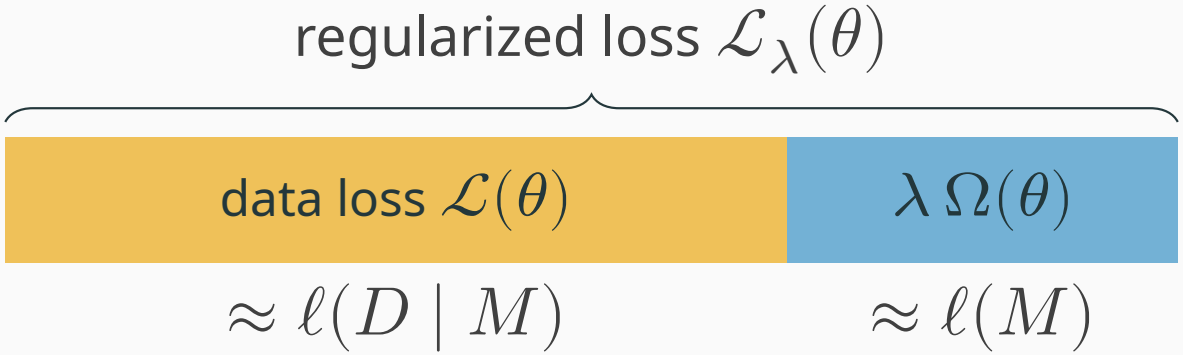
The **likelihood** of the data D is $p(D \mid \theta)$: their probability under the parameters θ.



- A higher likelihood means a shorter code for the data
- $\ell(D \mid M) = -\log_2 p(D \mid \theta)$: maximum likelihood = shortest code

Regularization: Paying for Complexity

Regularization adds a penalty to the loss:
$\mathcal{L}_\lambda(\theta) = \mathcal{L}(\theta) + \lambda\,\Omega(\theta)$.



- $\Omega(\theta)$ measures complexity, for example the sum of squared parameters
- $\lambda \geq 0$ sets its weight: $\lambda = 0$ is the plain fit
- The penalty plays the part of the model cost $\ell(M)$

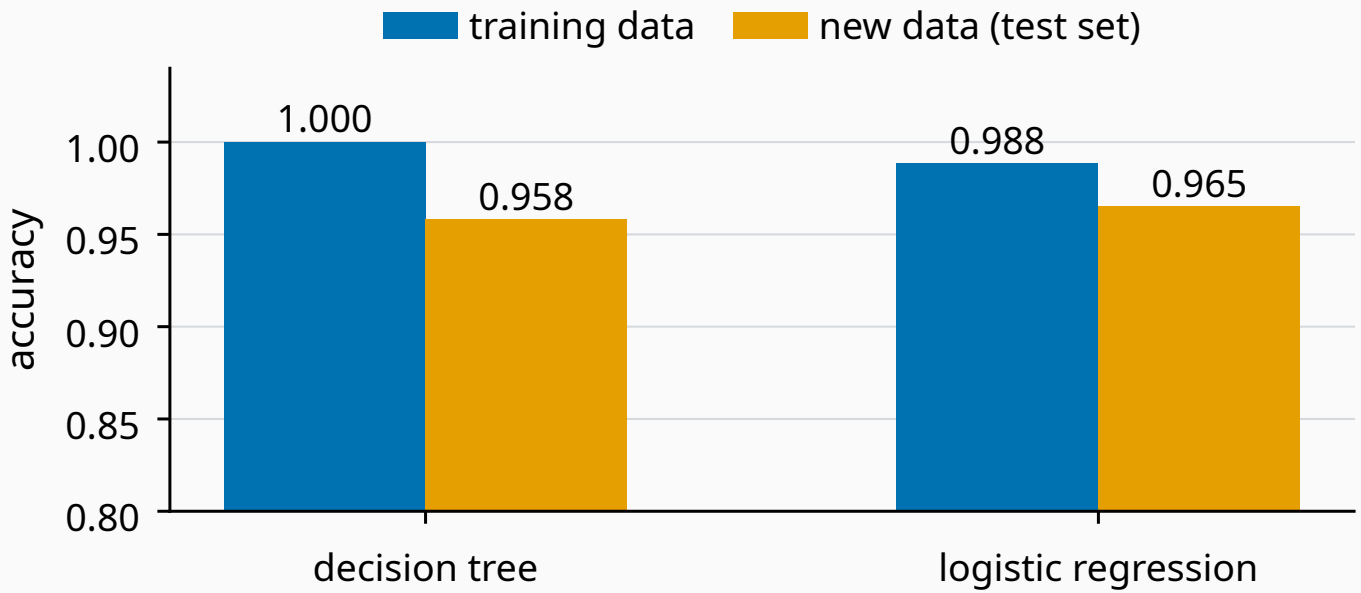
DEMO D3

`01_learning_as_compression.ipynb`

Evaluating a Model

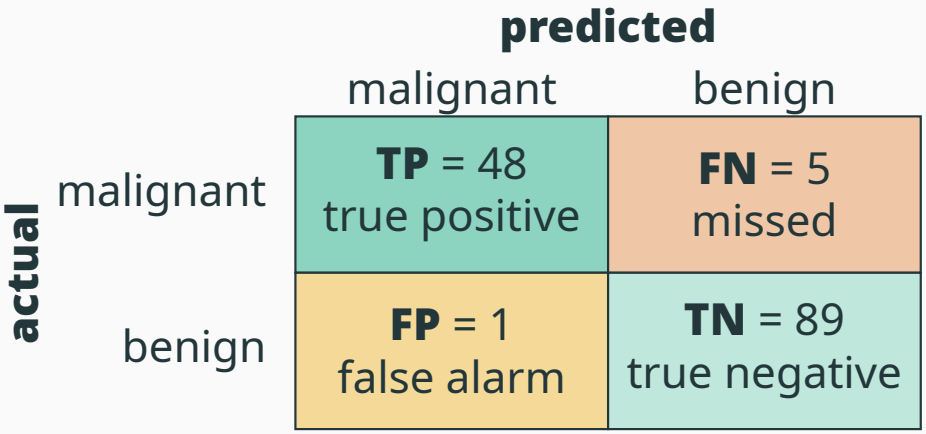
Training Loss Is Not the Goal

Example 2: 569 tumours, malignant ($y = 1$) or benign. Accuracy: share correct.



- Black boxes here: a decision tree (Class 06), logistic regression (Class 03)
- 143 unseen patients: tree 0.958, training 1.000; logistic regression 0.965, 0.988

The Confusion Matrix



- The tree on the 143 test patients; the positive class is malignant
- Each cell counts one kind of right or wrong decision
- Accuracy is $(48 + 89)/143 = 0.9580$

Precision, Recall, and F1

precision
flagged tumours that are malignant
48 of 49 = 0.9796

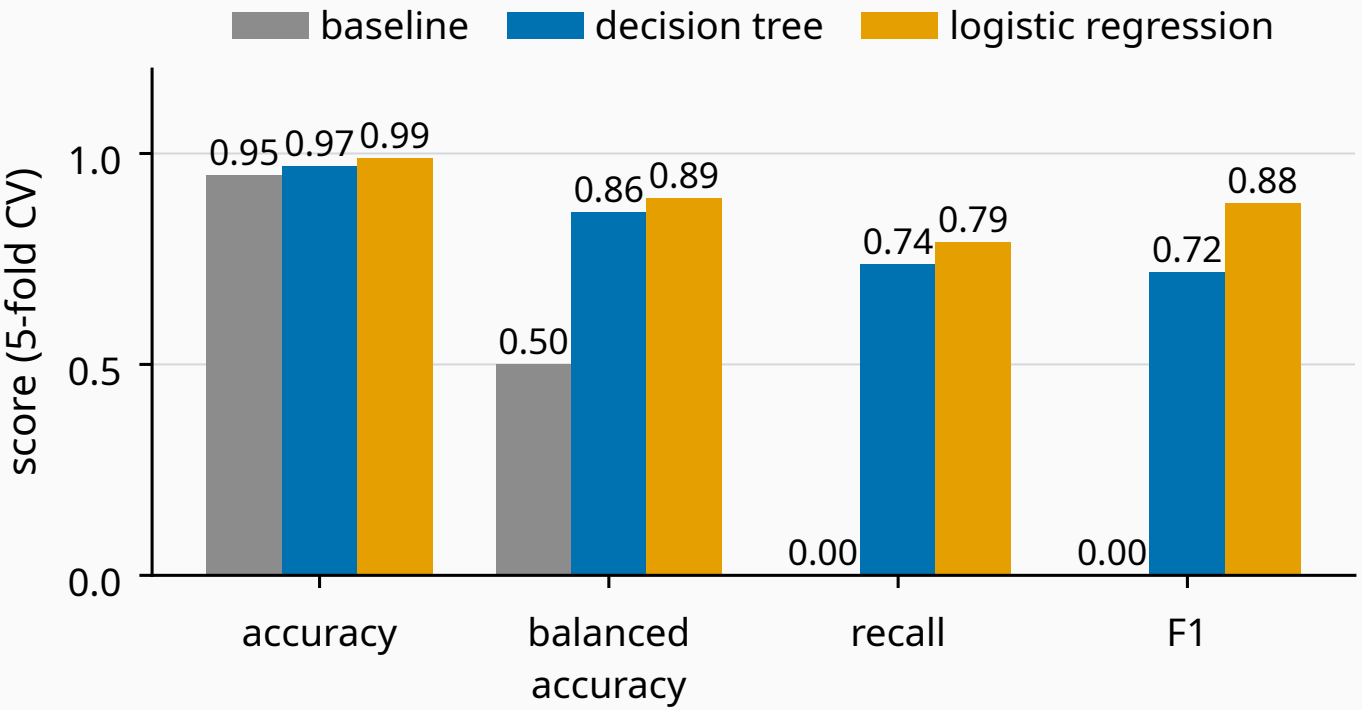
recall
malignant tumours that are found
48 of 53 = 0.9057

F1
balance of both
0.9412

- precision $= TP/(TP + FP)$
- recall $= TP/(TP + FN)$
- $F_1 = 2PR/(P + R)$, for precision P, recall R
- Screening needs recall; a spam filter needs precision

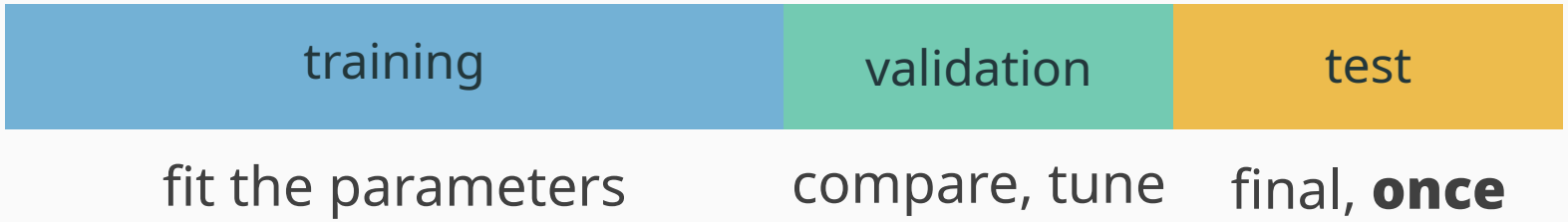
Baseline and the Accuracy Paradox

A **baseline** always predicts the most common class. A toy: breast cancer thinned to 5% malignant.



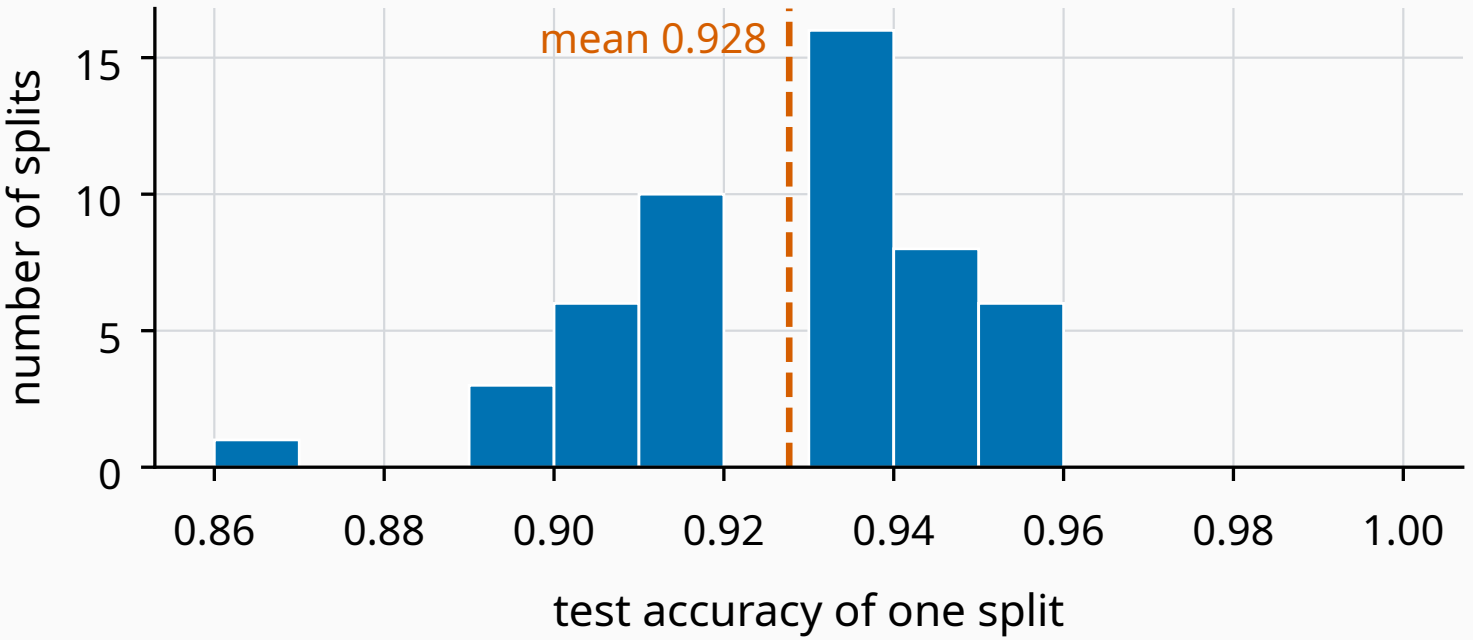
- Always benign: accuracy 0.949, recall 0.00
- Balanced accuracy, mean recall of both classes: 0.50
- Example 2: baseline 0.6294, tree 0.958

Hold Out Data: Training, Validation, and Test Sets



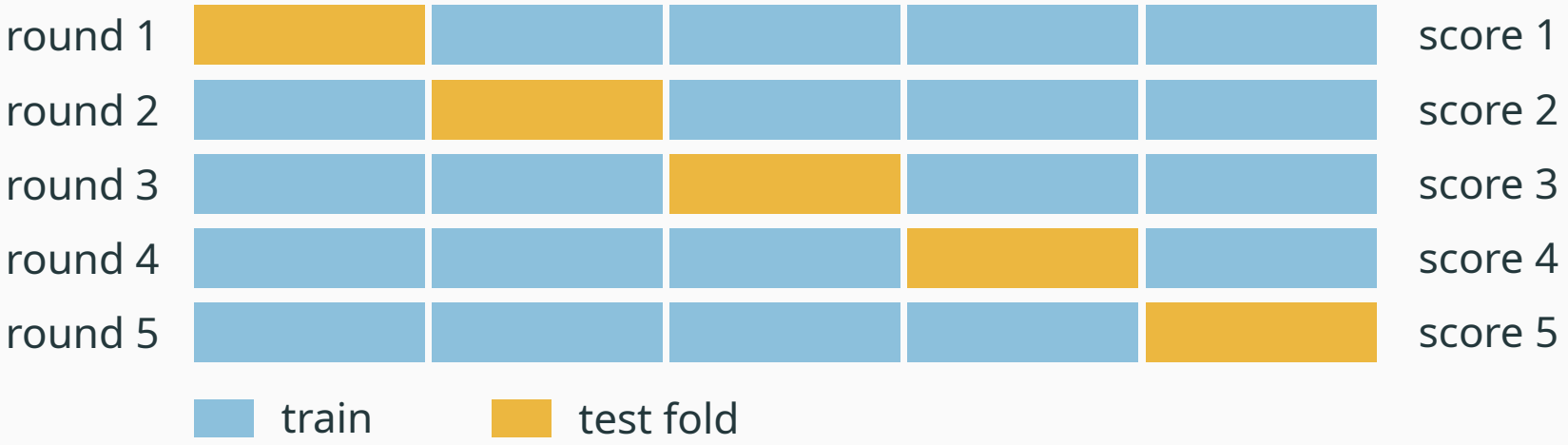
- Split the data before training
- Example 2: 426 patients train the model, 143 test it
- Every decision based on the test score makes it optimistic

One Split Is Not Enough



- Same tree, same data, 50 random splits: 0.860 to 0.958
- Mean 0.928, a spread of 9.8 points
- A single number hides this spread

k-Fold Cross-Validation and Its Variants



- Split the data into k parts (folds); here $k = 5$
- Each round tests one fold; every example is tested once
- Variants keep class shares (stratified), a patient's records together (grouped) or time order (temporal)

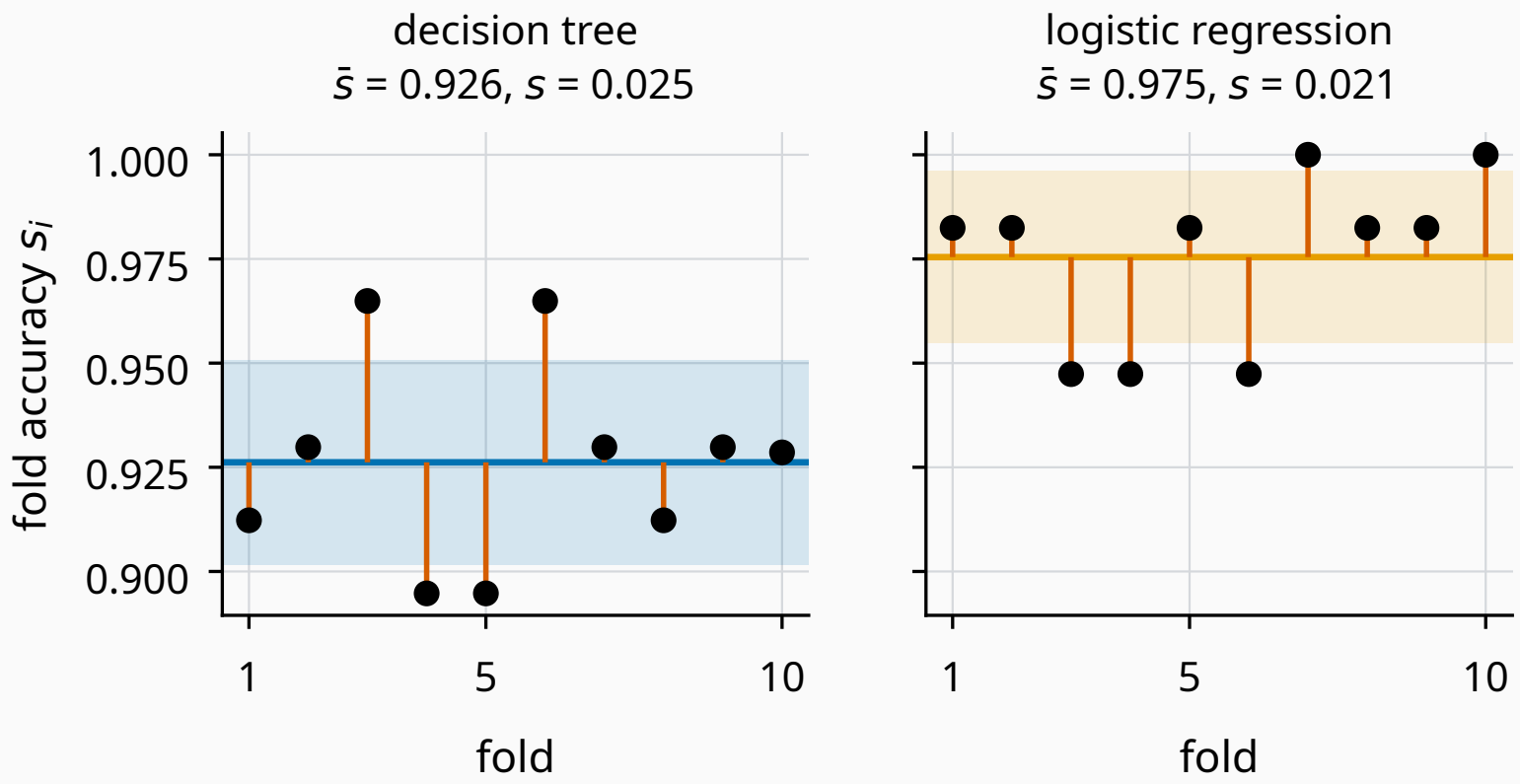
Notation N3a: Scores and Intervals

Example 2: decision tree, 10-fold cross-validation.

Symbol	Meaning	Example 2
k	number of folds	10
s_i	score (accuracy) on fold i	0.9123 for $i = 1$
$\bar{s}$	mean of the fold scores	0.9262
s	standard deviation of the fold scores	0.0245
SE	standard error of the mean, $s/\sqrt{k}$	0.0078
CI	95% confidence interval for the true accuracy	[0.9087, 0.9437]

Variance and Standard Deviation

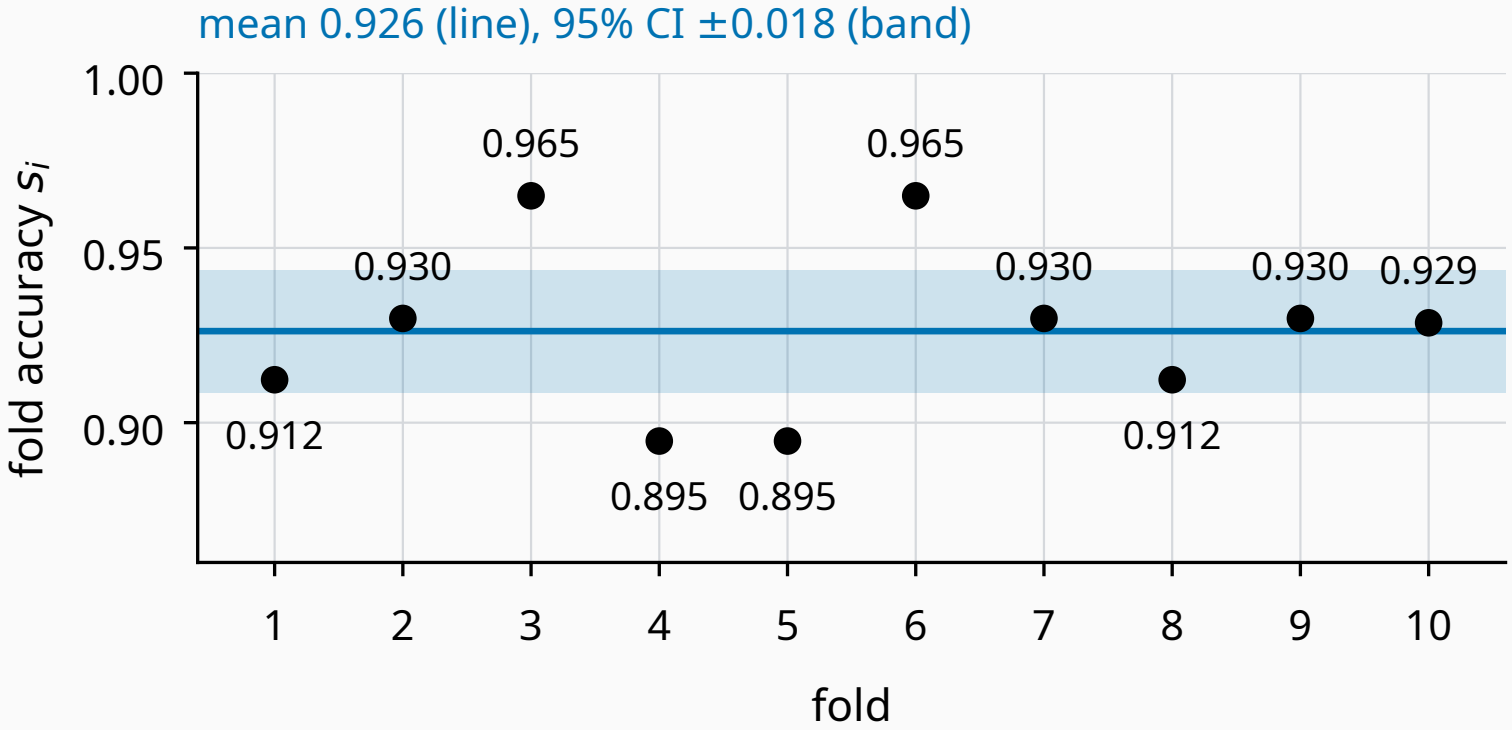
$$s^2 = \frac{1}{k-1}\sum_{i=1}^{k}(s_i - \bar{s})^2, s = \sqrt{s^2}.$$



- Divide by $k - 1$: the mean used up one score
- Simulated, true variance 1: $\div 5$: 0.801, $\div 4$: 1.001

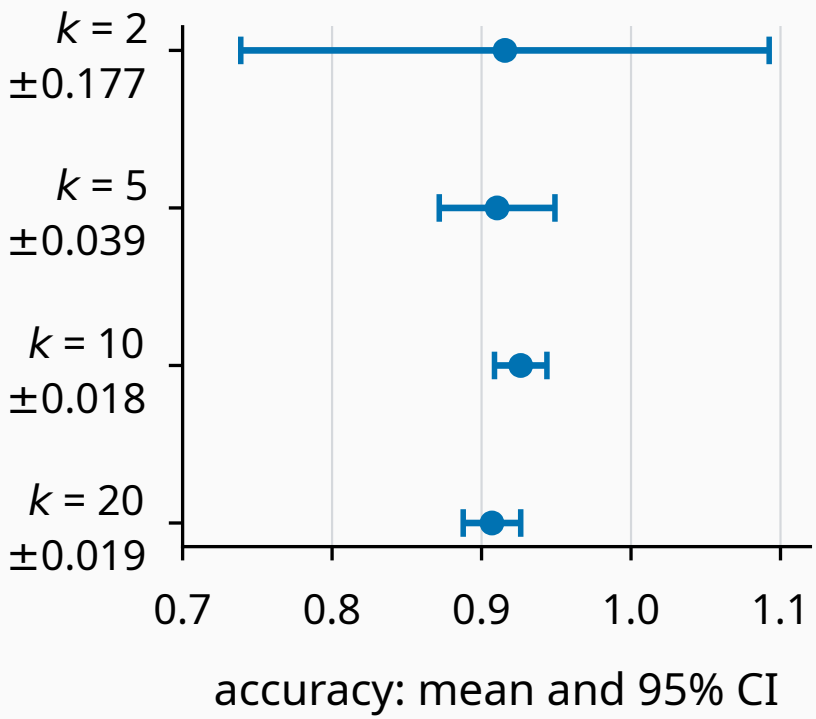
The Confidence Interval

SE $= s/\sqrt{k}$ and CI $= \bar{s} \pm t_{0.975,\,k-1}$ SE.



- $t_{0.975,\,k-1}$: 97.5% point of Student's t, $k-1$ degrees of freedom
- Degrees of freedom: scores free once the mean is fixed

Worked Example: Confidence Interval

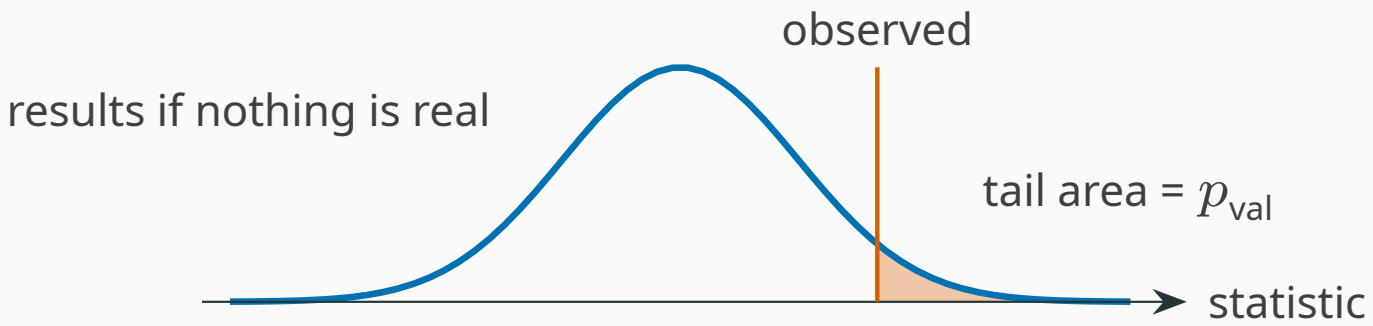


accuracy: mean and 95% CI

1. $\bar{s} = 0.9262$, $s = 0.0245$
2. SE $= 0.0078$
3. $t_{0.975,\,9} = 2.262$
4. Half-width $= 0.0175$

- Report 0.9262 ± 0.0175: the interval is [0.9087, 0.9437]
- 95% of such intervals hold the truth
- Limit: folds share training data: too narrow

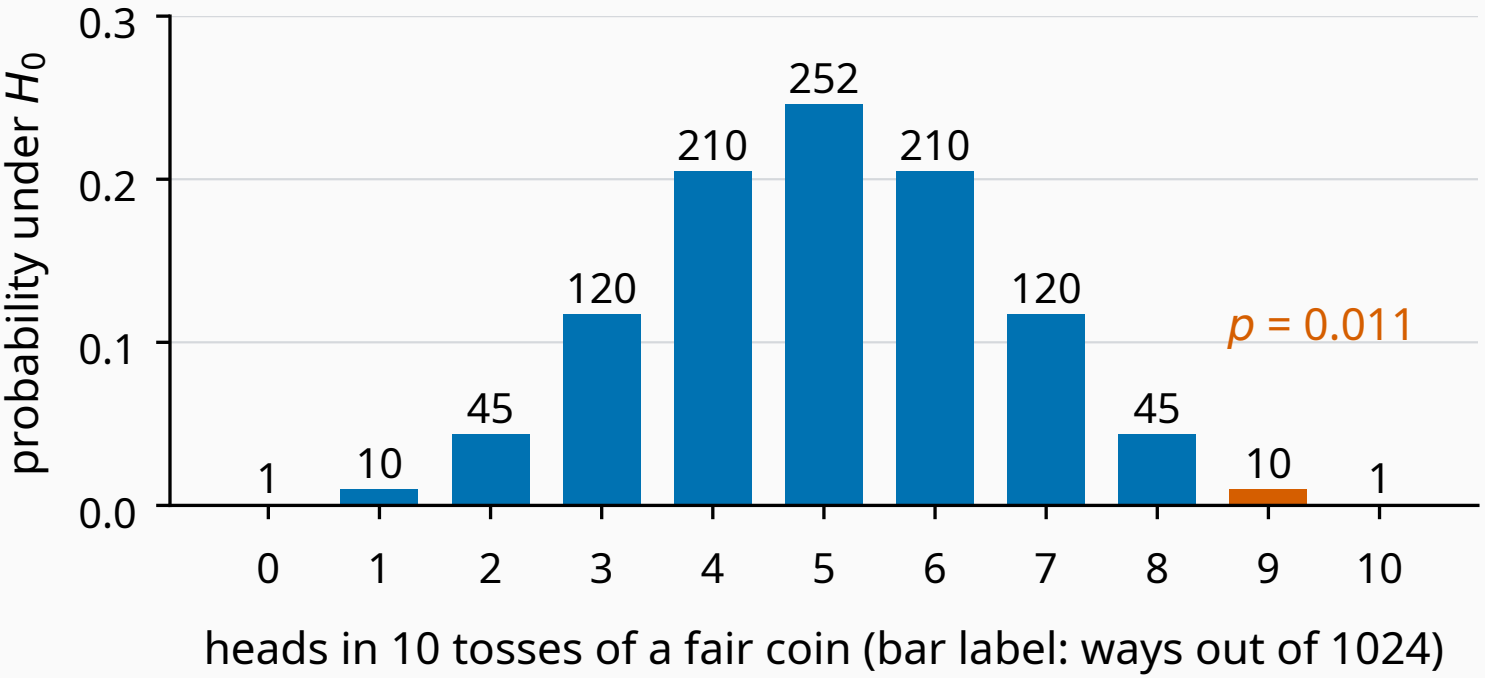
Is the Difference Real? Hypothesis Testing



- Null hypothesis H_0: nothing real, only luck
- p-value: chance of a result this extreme if H_0 holds
- Reject H_0 when $p_{\text{val}} < \alpha = 0.05$, the significance level
- It is not the chance that H_0 is true

Worked Example: a Coin and a Classifier

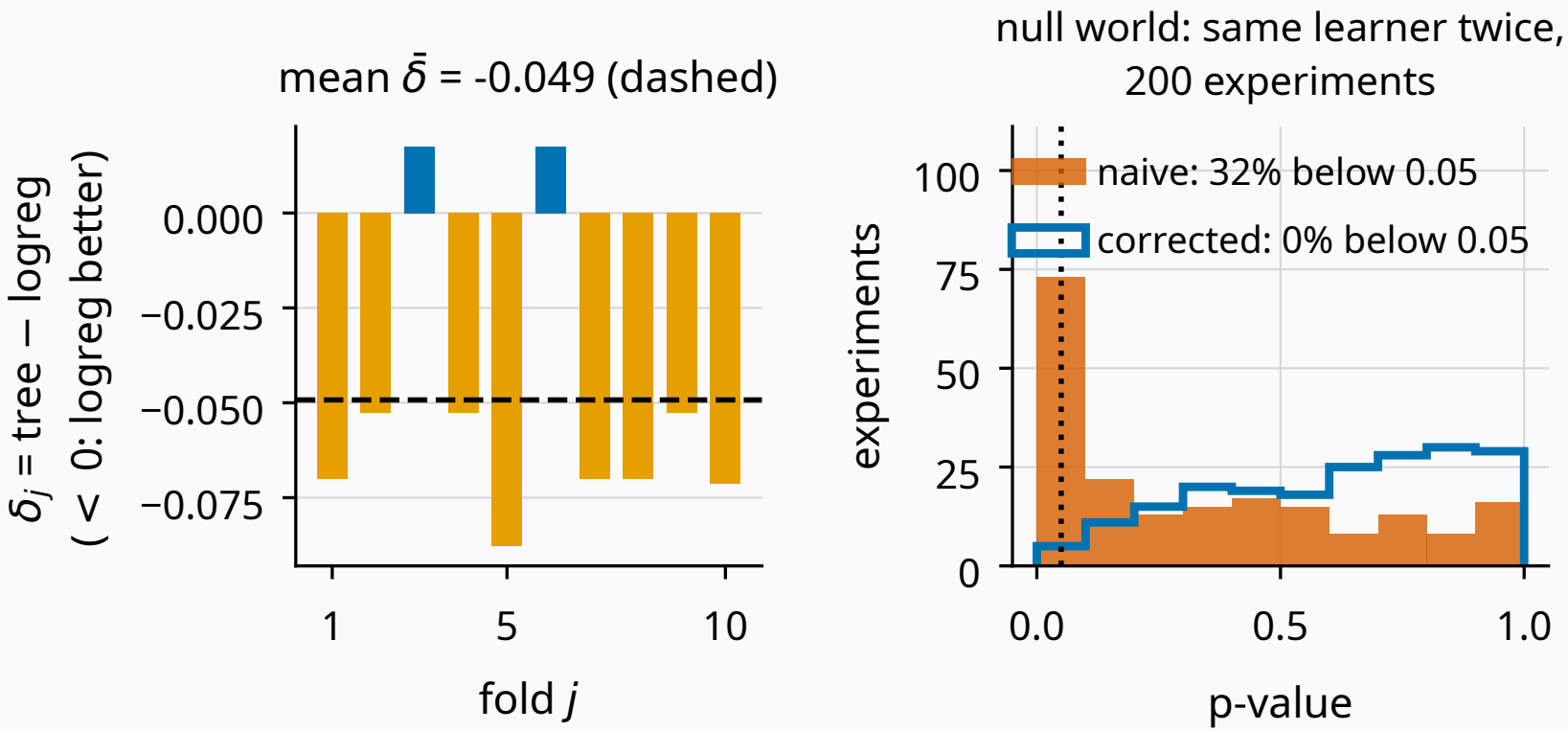
Null hypothesis: a fair coin. We see 9 heads in 10 tosses.



- 11 of 1024 outcomes are this extreme: $p_{\text{val}} = 0.01074$
- The tree gets 137 of 143 test patients right; null: the baseline's 0.6268
- Its p-value is 4.964e-21: not luck

Comparing Two Models: Pair the Scores

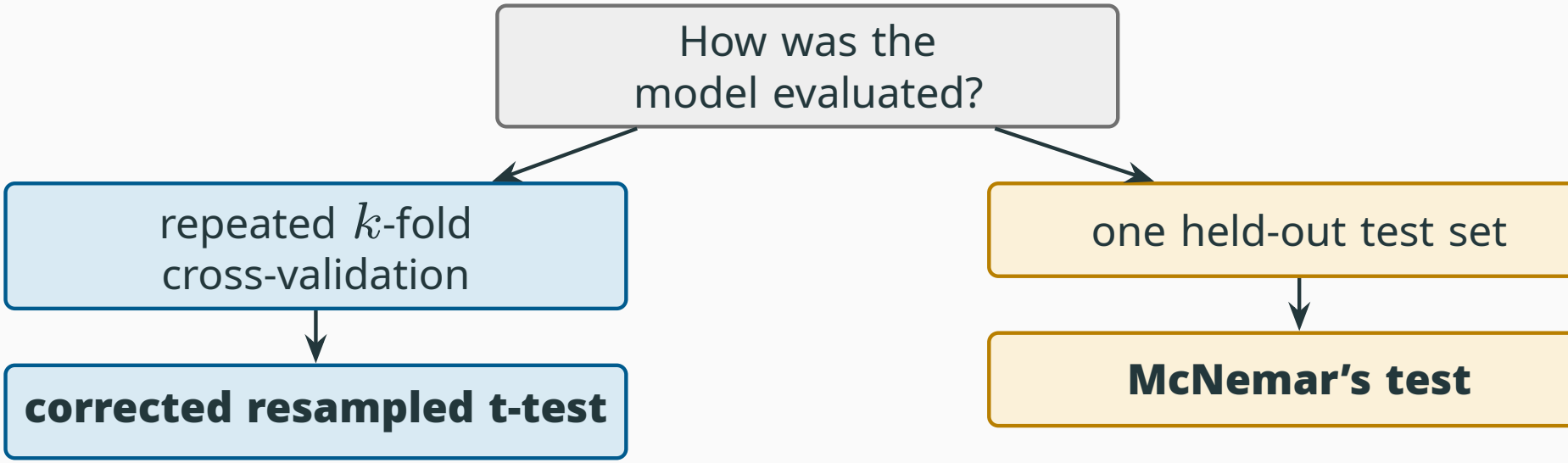
$\delta_j = s_j^A - s_j^B$: A tree, B logistic regression.



power on a real difference: naive 96.5%, corrected 59.5% (conservative)

- Mean difference $\bar{\delta} = -0.0492$: B wins in 8 of 10 folds
- False positive rate: plain t-test 31.5%, corrected 0.0%

Which Test When?



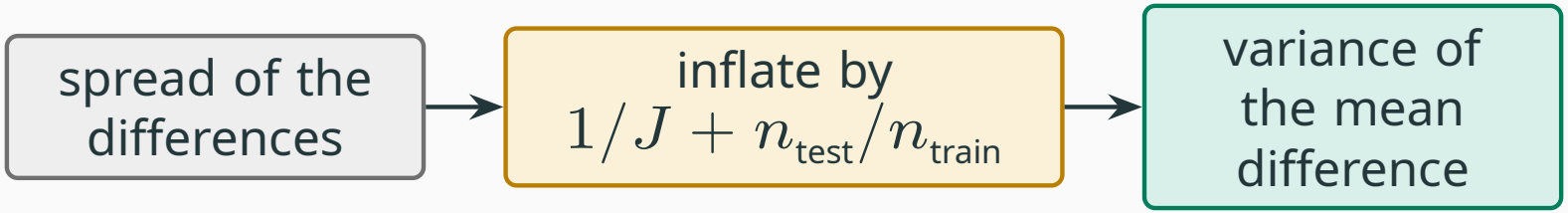
- One score per fold or split: compare the scores
- One prediction per patient: compare who is right

Notation N3b: Tests

Symbol	Meaning	Example 2
J	number of scores	50 (5 $\times$ 10-fold)
$\bar{\delta}$, s_δ	mean score difference, its standard deviation	-0.0520, 0.0327
n_{train}, n_{test}	examples to train, to test in one split	ratio 0.1111
n_{11}, n_{00}	both models right, both wrong	520, 7
n_{10}, n_{01}	only A right, only B right	7, 35
N	disagreements, $n_{10} + n_{01}$	42
T, χ^2	t statistic, chi-squared statistic	-4.39, 18.67
p_{val}, α	p-value, significance level	6.01e-05, 0.05

The Corrected Resampled t-Test

Is the mean difference of the J scores real?



$$T = \bar{\delta} \Big/ \sqrt{\left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}}\right) s_\delta^2}$$ (by Nadeau and Bengio 2003)

- Example 2, $J = 50$: $\bar{\delta} = -0.0520$, $s_\delta = 0.0327$
- $n_{\text{test}}/n_{\text{train}} = 0.1111$
- $T = -4.39$, p-value 6.01e-05 (naive $T = -11.24$); conservative: power 59.5% against 96.5%

McNemar's Test: the Statistic

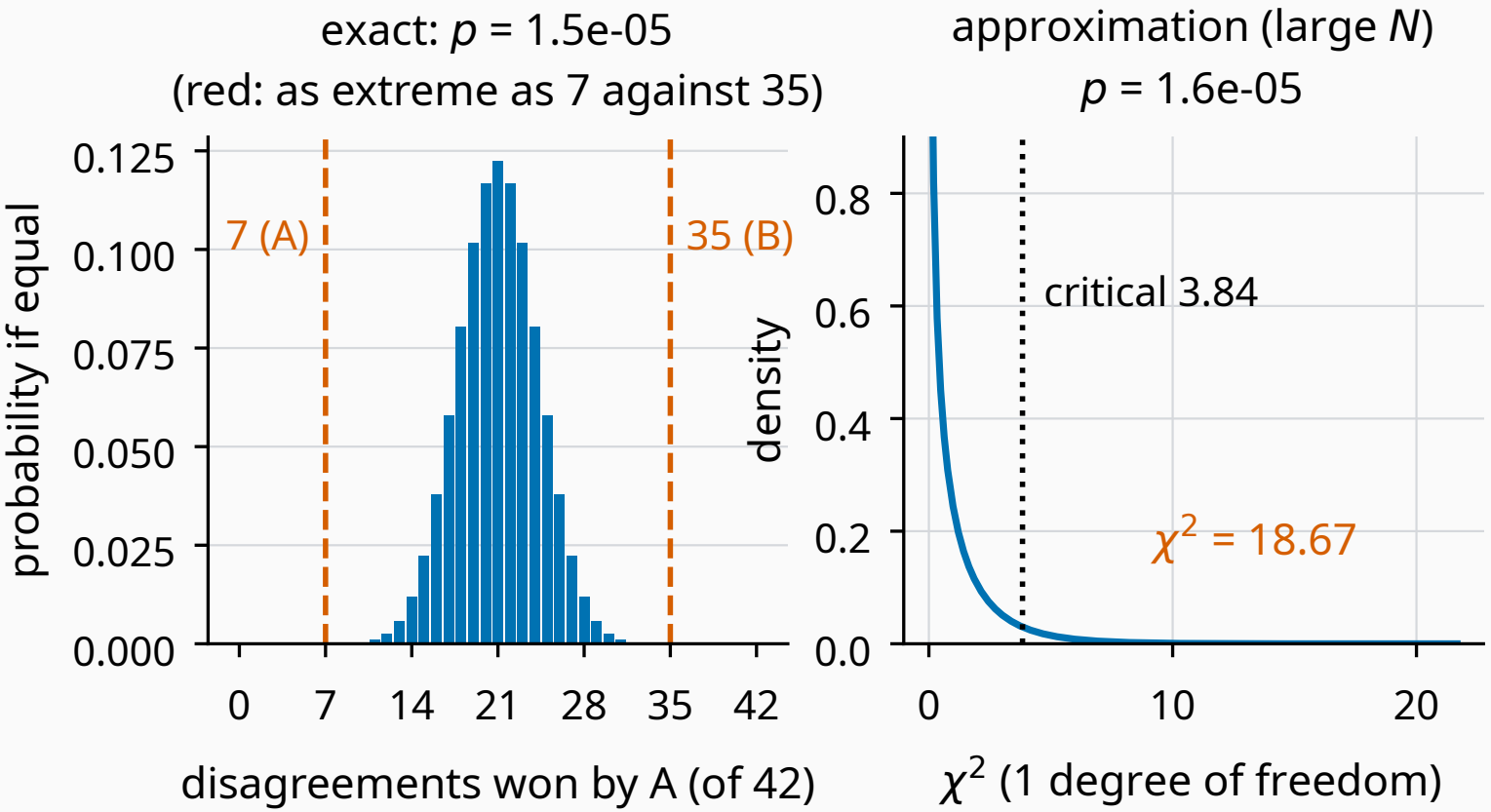
Do A and B err on different patients? 569 out-of-fold predictions.

	B right	B wrong
A right	$n_{11} = 520$	$n_{10} = 7$
A wrong	$n_{01} = 35$	$n_{00} = 7$

$$\chi^2 = \frac{(n_{10} - n_{01})^2}{n_{10} + n_{01}}$$ (by McNemar 1947)

- Only the $N = 42$ disagreements count: if A = B, fair coin tosses
- $\chi^2 = 28^2/42 = 18.67$, p-value 1.56e-05
- Edwards 1948, -1 in the square: 17.36, 3.10e-05

McNemar's Test: Meaning and Limit



- A wins 7, B wins 35: B is better; exact p-value 1.51e-05
- Limit: for $N < 25$ use the exact test, not χ^2
- Test set: table 133, 4, 5, 1, $N = 9$, exact p-value 1.0

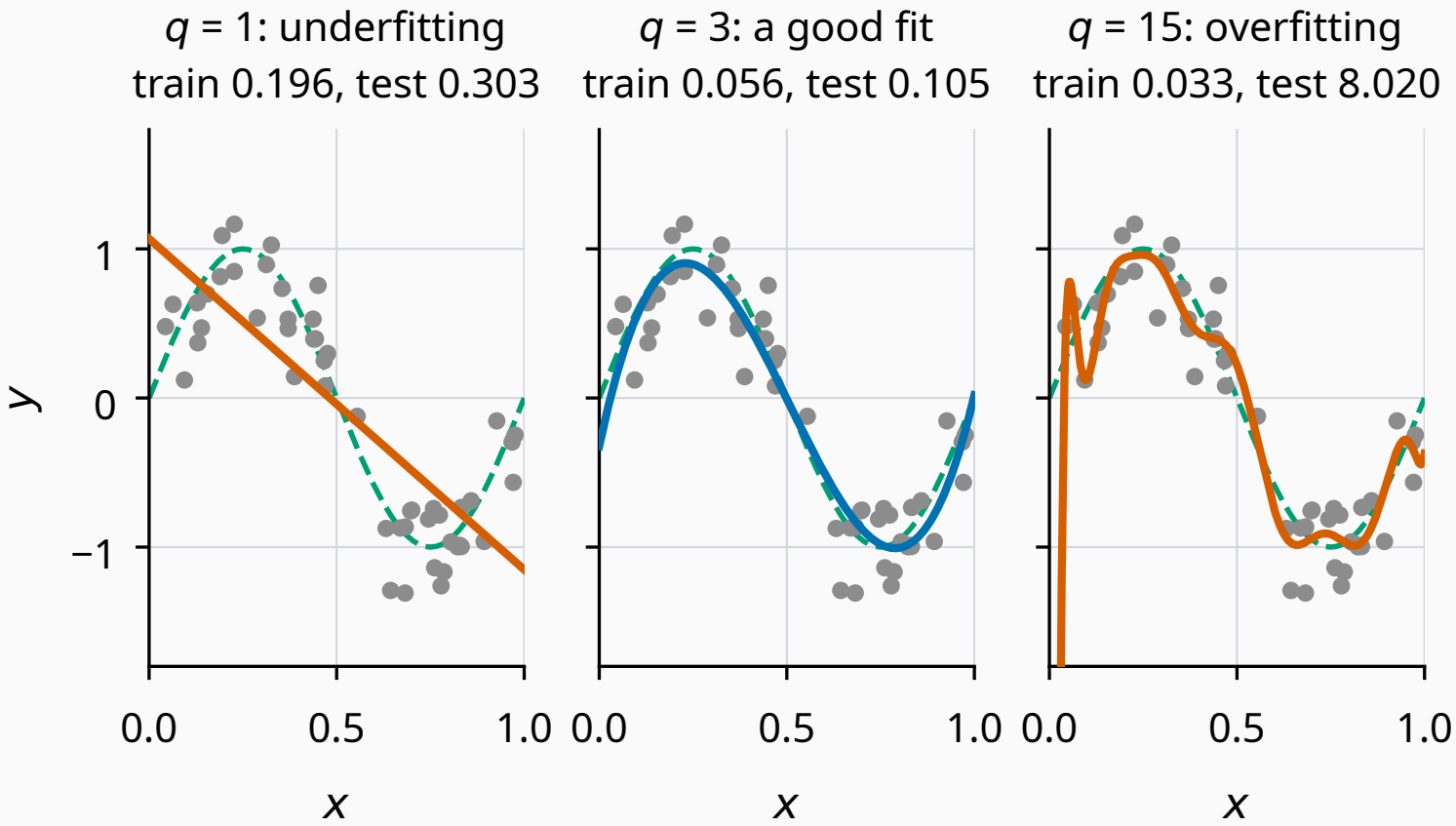
DEMO D4

`02_evaluation_validation.ipynb`

D4a–D4d live; D4e–D4g read in the lab

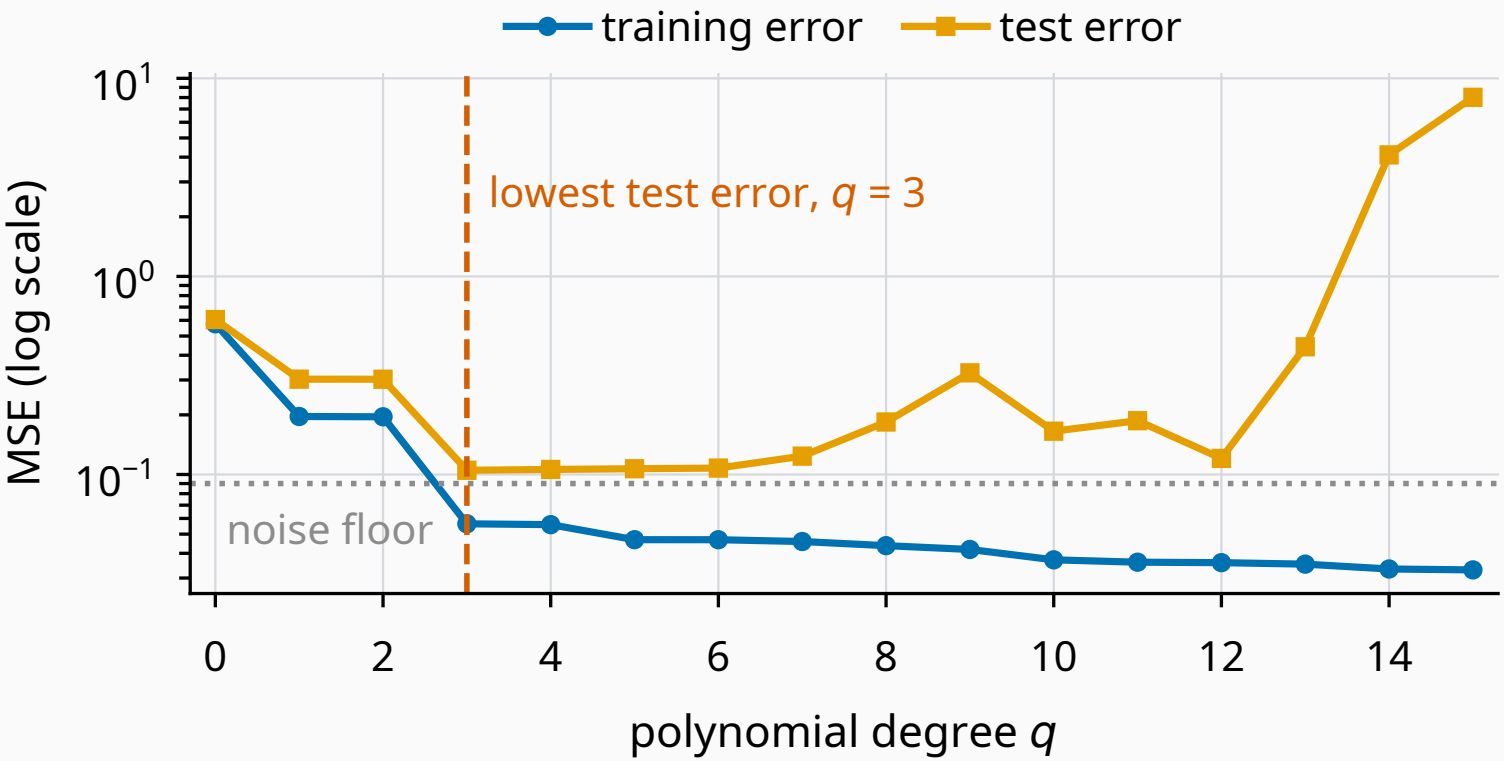
Why Models Fail

Underfitting, a Good Fit, Overfitting



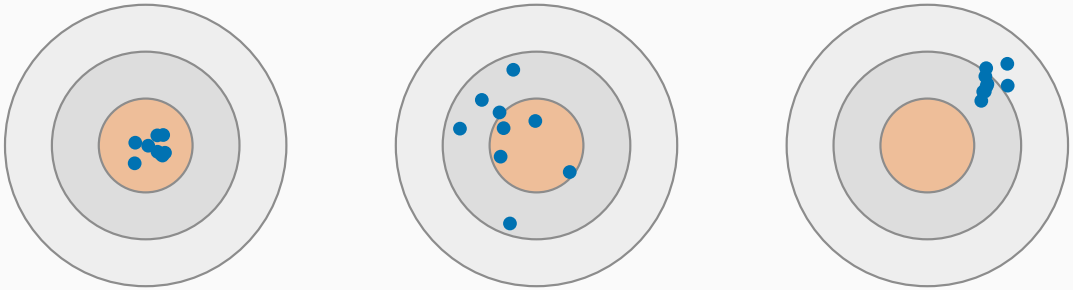
- Underfitting: a line misses the wave, test 0.3028
- Overfitting: degree 15 copies the sample, test 8.0205

Error Versus Model Complexity



- Training error always falls as q grows
- Test error is lowest at $q = 3$: 0.1050
- Beyond it the model fits noise: test error rises

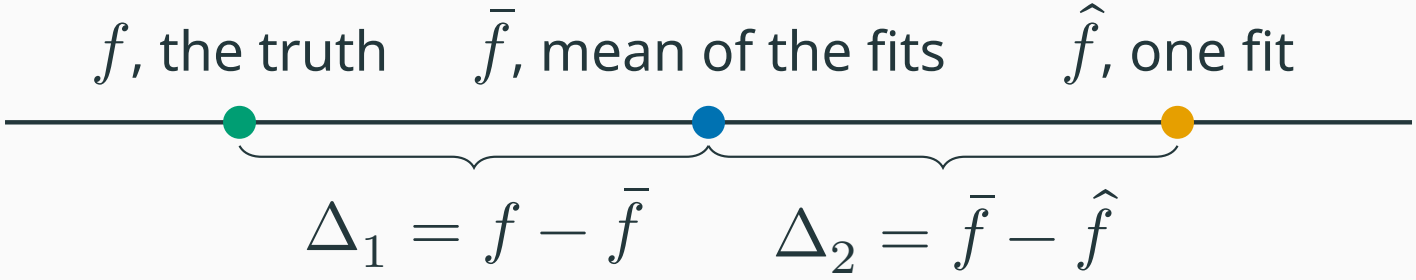
Bias and Variance



low bias, low variance | low bias, high variance | high bias, low variance

- Each dot is one fit on a new training set; the bullseye is f
- **Bias**: error left after averaging many fits, $\bar{f} - f$
- **Variance**: how much the fit changes with the sample
- Example 1, 200 training sets: line $q = 1$, bias^2 0.2000, variance 0.0157; $q = 9$: 0.0006, 0.1298

The Bias-Variance Decomposition

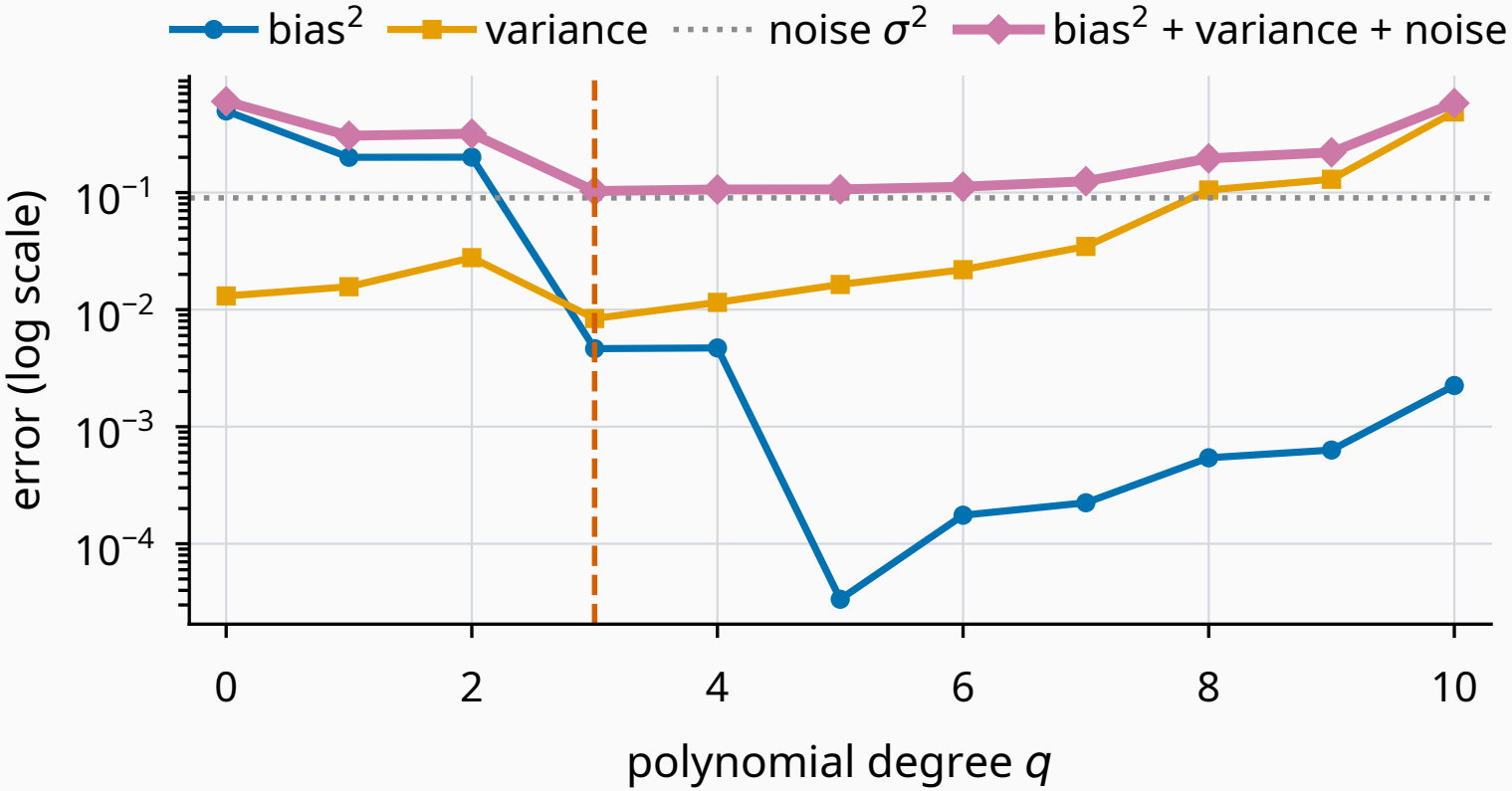


$$
\begin{aligned}
(y - \hat{f})^2 &= (\Delta_1 + \Delta_2 + \varepsilon)^2 && \text{(substitute)} \\
&= \Delta_1^2 + \Delta_2^2 + \varepsilon^2 + 2(\Delta_1\Delta_2 + \Delta_1\varepsilon + \Delta_2\varepsilon) && \text{(expand)} \\
\mathbb{E}[(y - \hat{f})^2] &= \Delta_1^2 + \mathbb{E}[\Delta_2^2] + \sigma^2 && \text{(assume)} \\
&= \text{bias}^2 + \text{variance} + \sigma^2 && \text{(definition)}
\end{aligned}
$$

Assumptions behind row 3 (they kill the cross terms):

- $\mathbb{E}[\varepsilon] = 0$; ε is independent of $\hat{f}$
- $\mathbb{E}[\hat{f}] = \bar{f}$, so $\mathbb{E}[\Delta_2] = 0$; f, $\bar{f}$ are fixed

The Trade-Off and What to Do About It



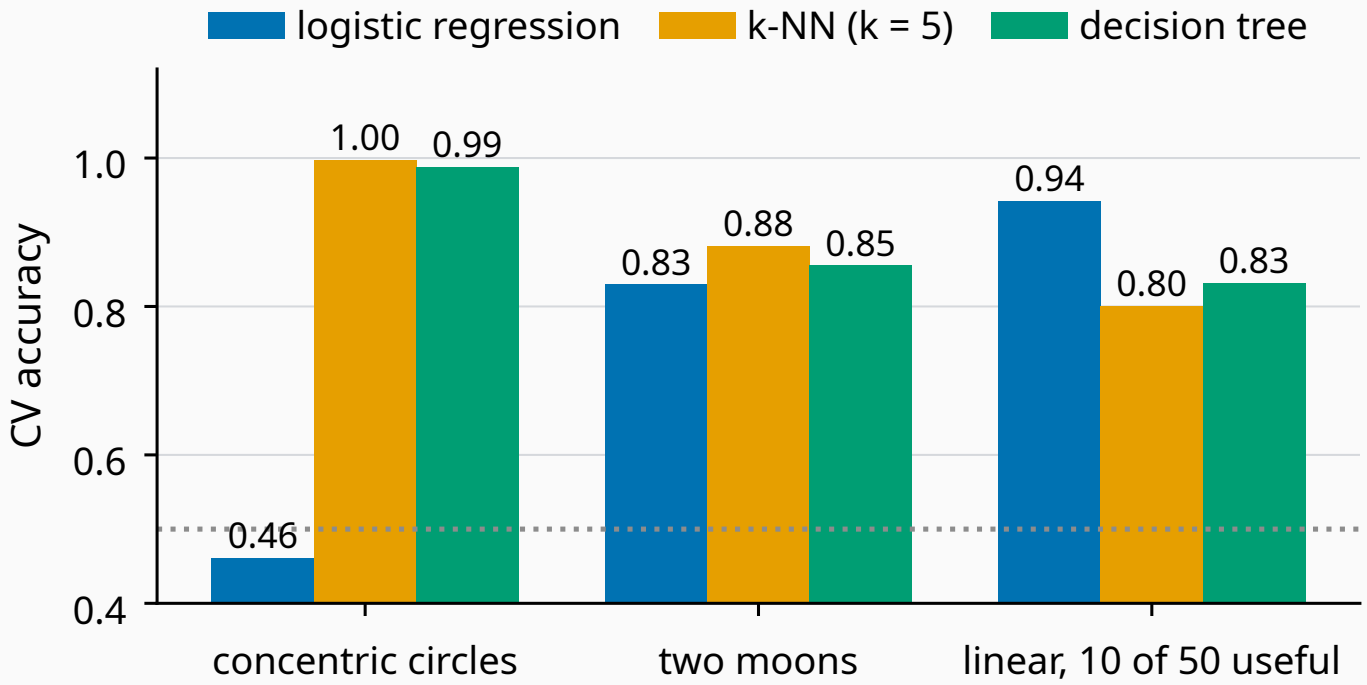
- More complexity lowers bias and raises variance
- Lowest sum at $q = 3$: 0.0046 + 0.0084 + 0.09

DEMO D5

`03_pitfalls.ipynb`

No Free Lunch and Inductive Bias

Over all possible problems, no learner beats another (Wolpert, 1996).



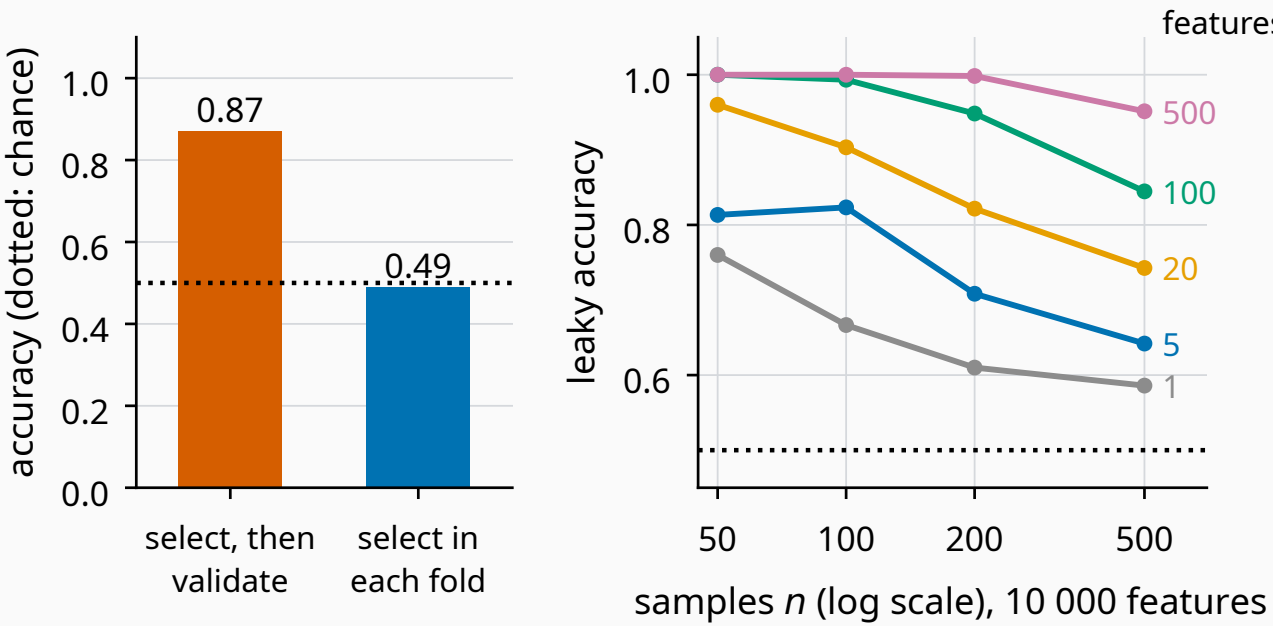
- Each model assumes something about the pattern
- Circles: logistic regression 0.461, k-NN 0.997; k-NN copies the nearest neighbours (Class 06)

DEMO D6

`03_pitfalls.ipynb`

Data Leakage and the Right Pipeline

Leakage: information from outside the training data reaches the model.



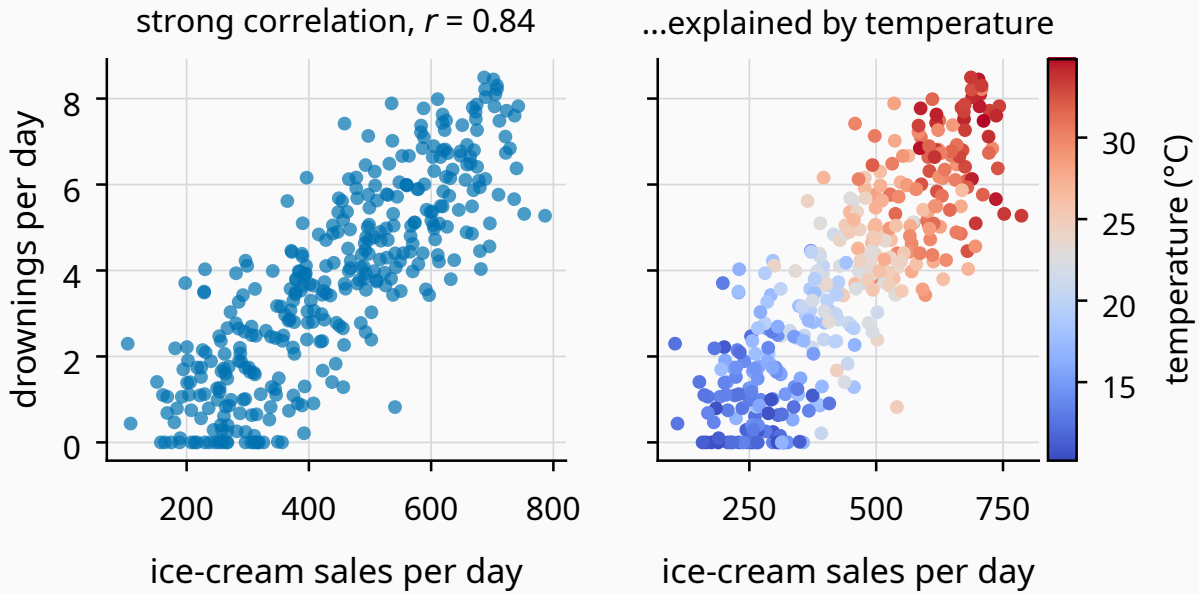
- Random labels, 100 patients, 10 000 features: select the 20 best
- Selecting before the split: 0.870; inside each fold, the right way: 0.490

DEMO D7

03_pitfalls.ipynb

Correlation and the Ice-Cream Example

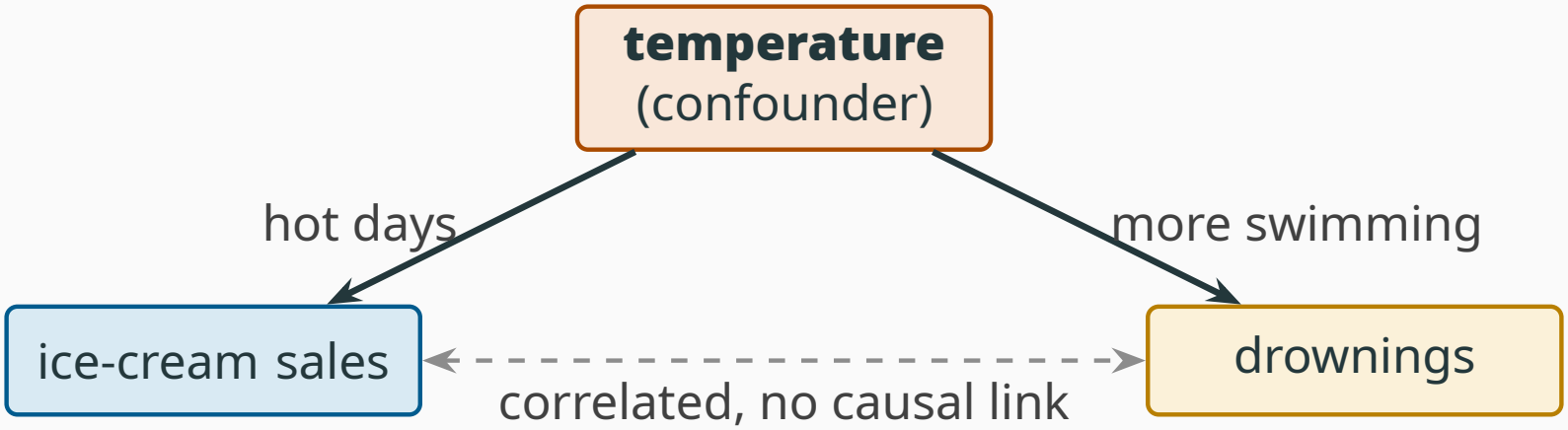
r: how two variables move together; $\bar{x}$, $\bar{y}$ are means.



$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2 \sum_i (y_i - \bar{y})^2}} \quad \text{(definition)}$$

- Ice-cream sales and drownings: $r = 0.838$

The Hidden Common Cause



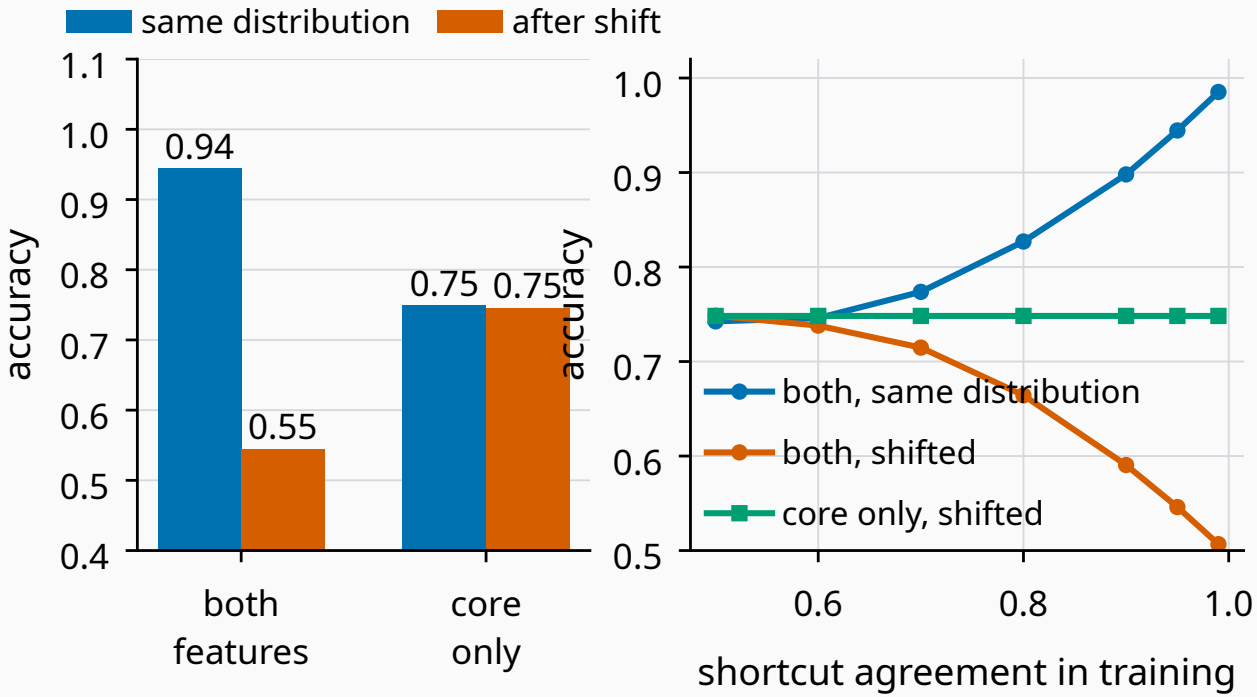
- A confounder causes both variables
- Ice-cream weight alone: $+1.2107$ drownings per 100 sales
- With temperature added, a regression with two inputs (Class 03): -0.0186, about zero
- Halving sales: the model predicts 384.2 drownings, reality gives 1351.2

DEMO D8

03_pitfalls.ipynb

Shortcuts and Distribution Shift

A **shortcut** predicts in training, not in use. A sample that misses a group shifts the data too.



- Both features 0.945, then 0.545; core only 0.750, 0.746
- Shifts: $p(x)$ covariate, $p(y)$ label, $p(y \mid x)$ concept

DEMO D9

`03_pitfalls.ipynb`

Five Questions to Ask of Any Result

Baseline? better than the simplest

Split? tested on unseen data

Leakage? nothing from the test

Cause? more than correlation

Shift? same data in use

- Ask all five before you trust a number

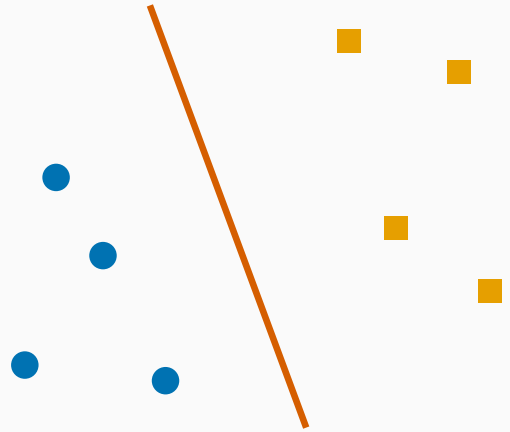
Kinds of Learning

Where the Learning Signal Comes From

Type	Tumour images with …	Learning signal	Classes
Supervised	a diagnosis each	the diagnosis	03–07
Unsupervised	no diagnosis	structure of the data	09–10
Semi-supervised	a few diagnoses	labels + structure	11
Self-supervised	no diagnosis	targets built from the data	11–12
Reinforcement	treatment outcomes	rewards	13

- Types differ in what tells the model it does well
- Self-supervised: hide part of the data, predict it

Supervised and Unsupervised Learning



supervised:
labels, a boundary

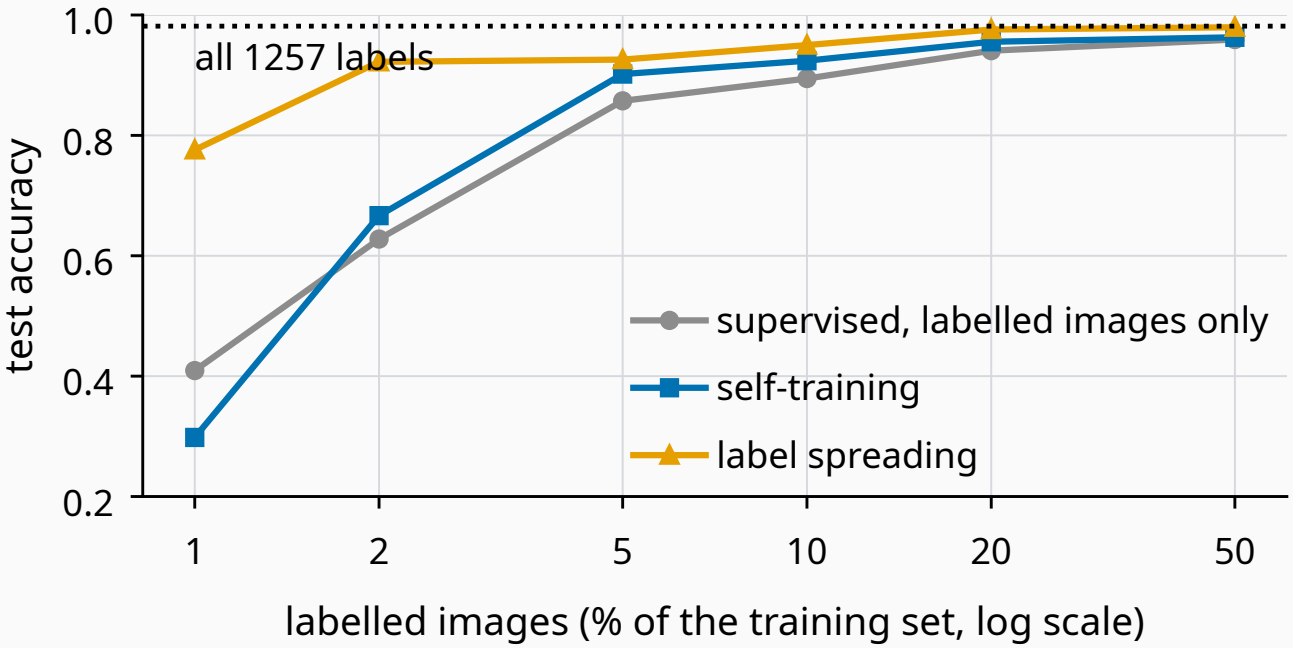


unsupervised:
no labels, groups

- Supervised: every example has its answer y
- Classification: y a category; regression: y a number
- Unsupervised: only inputs x, no labels
- Clustering: Class 10; sources, reduction: Class 09

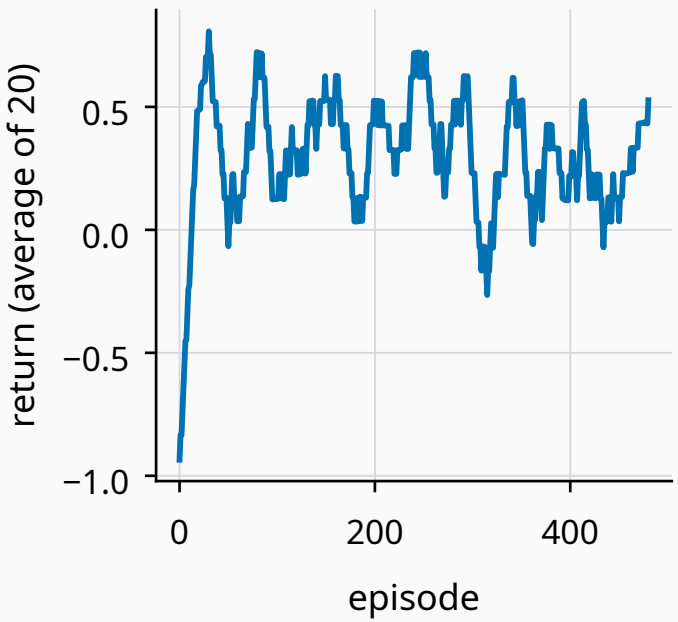
Semi-Supervised Learning

Few labels, many unlabelled examples.

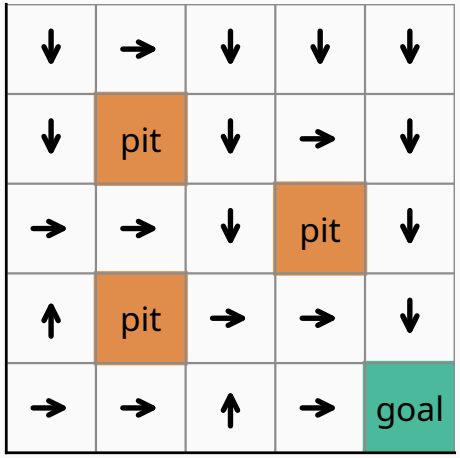


- Cluster assumption: dense groups share a label
- Self-training labels what it is sure of; label spreading passes labels to neighbours
- At 10% labels: 0.894, 0.924, 0.950 (supervised, self-training, spreading)

Reinforcement Learning



learned policy: best action per cell



Symbol	Meaning
S_t, a_t, R_{t+1}	state, action, reward at time t
policy	rule that picks the action
episode, return	one run from start to end, its summed reward
pit, goal	cells that end an episode: penalty, reward

DEMO D10

04_taxonomies.ipynb

Wrap-Up

Key Takeaways, the Lab and Class 02

1. Fit
minimize a loss over the parameters

2. Compression
keep the pattern, drop the noise

3. Evaluation
splits, intervals, tests

4. Trade-off
bias against variance

5. Pitfalls
leakage and shift

6. Learning types
the learning signal

- Lab, 60 minutes: Parts A to F in class, Part G at home
- Class 02: learning as search over parameters, blind and gradient-based

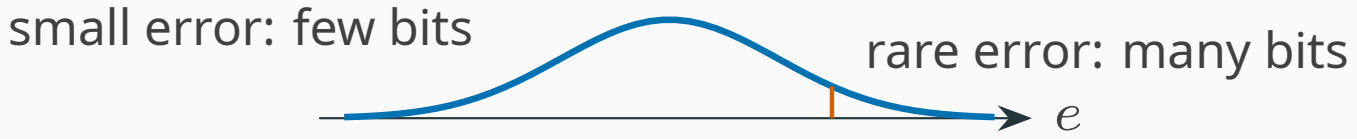
References

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- J. Rissanen, “Modeling by Shortest Data Description”, *Automatica*, 1978
- D. H. Wolpert, “The Lack of A Priori Distinctions Between Learning Algorithms”, 1996
- T. M. Mitchell, *Machine Learning*, 1997
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Appendix

Where BIC Comes From: the Error Cost

Each Gaussian error e_i has a surprise $h(e_i)$; the n surprises add up.



$$
\begin{aligned}
h(e) &= -\log_2 p(e) && \text{(definition of surprise)} \\
&= -\log_2 \frac{\exp\left(-e^2/(2\sigma^2)\right)}{\sqrt{2\pi\sigma^2}} && \text{(by the Gaussian density)} \\
&= \tfrac{1}{2}\log_2(2\pi\sigma^2) + \frac{e^2}{2\sigma^2 \ln 2} && \text{(log of a product)}
\end{aligned}
$$

- Small errors have a high density: few bits

Where BIC Comes From: Sum Over the Examples

So far: surprise of one error. **Goal:** code length of all errors.

$$\begin{aligned}
\ell(D \mid M) &= \sum_{i=1}^{n} h(e_i) && \text{(definition)} \\
&= \tfrac{n}{2} \log_2(2\pi\sigma^2) + \frac{\sum_i e_i^2}{2\sigma^2 \ln 2} && \text{(linearity of the sum)} \\
&= \tfrac{n}{2} \log_2(2\pi\sigma^2) + \frac{n \,\mathrm{MSE}}{2\sigma^2 \ln 2} && \text{(definition of MSE)}
\end{aligned}$$

Where BIC Comes From: the Constant Drops Out

So far: code length of the errors. **Goal:** the BIC form.

$$\begin{aligned}
\ell(D \mid M) &= \tfrac{n}{2} \log_2(2\pi\sigma^2) + \frac{n \,\mathrm{MSE}}{2\sigma^2 \ln 2} && \text{(definition)} \\
&= \tfrac{n}{2} \log_2(2\pi \,\mathrm{MSE}) + \tfrac{n}{2 \ln 2} && \text{(substitute)} \\
&= \tfrac{n}{2} \log_2 \mathrm{MSE} + c && \text{(definition of } c\text{)}
\end{aligned}$$

- $\sigma^2 = \mathrm{MSE}$: the fitted variance; c is shared by all models

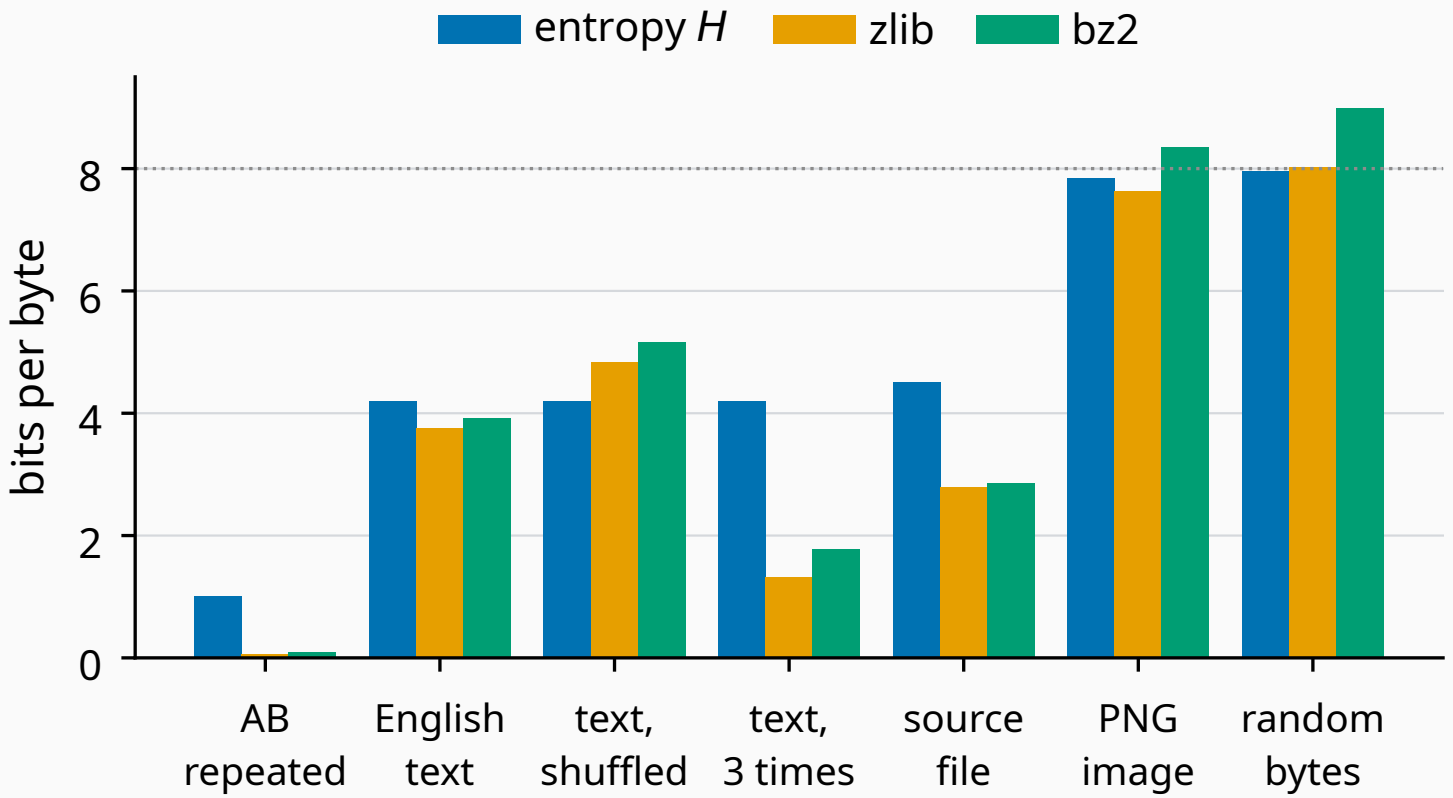
Kolmogorov Complexity

$K(u)$ is the length of the shortest program that prints the string u.

`ABABABABABABABABABABABAB`	$\rightarrow$ "AB $\times$ 12"
`THE CAT SAT ON THE MAT.`	$\rightarrow$ some savings
`Q7#pZ2!mK9vX@4rT&1wL8`	$\rightarrow$ no shorter description

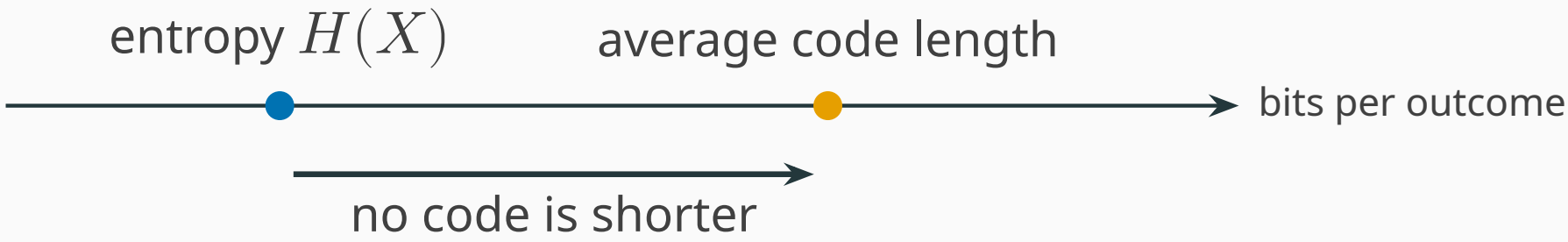
- Structure has a short description; randomness has none
- Pure noise leaves nothing to learn

Kolmogorov Complexity Versus Compression



- Repeating two letters 2000 times: 18 characters of code
- $K(u)$ is not computable; a compressor such as zlib gives an upper bound

The Source Coding Theorem



$$\mathbb{E}\big[\ell(X)\big] \geq H(X) \quad \text{(by Shannon 1948)}$$

- $\ell(X)$ counts the bits of the code for outcome X
- A trusted statement: the proof is in every information-theory text